Optimizing Institutional Food Systems via Transactional Data: A Residential University Case Study

by

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Abstract

Food wastage in institutional dining facilities is a multidimensional problem with moral, financial, sanitary, and operational implications. This study adopts a Data Warehousing and Business Intelligence (DWBI) framework to analyze historical mess expenditure data from a mess database as a proxy for consumption behavior. Using financial transactions from August 2025 - December 2025 academic period, we quantify patterns, identify inefficiencies, and propose targeted interventions. The analysis reveals structural inefficiencies in procurement and vendor dependency through temporal trend analysis, Benford's Law validation, and network analysis of the mess-vendor financial system.Frequent pattern mining identified 123 strong association rules revealing four critical vendor dependencies that collectively account for over 43% of all procurement activity.

Chapter 1

Introduction and Problem Motivation

From a financial perspective, institutional mess facilities operate on constrained budgets, often funded collectively by students or subsidies. Persistent wastage directly translates to avoidable expenditure. Sanitarily, excess food increases the burden on waste management systems and can attract pests or lead to spoilage-related hazards. Operationally, inaccurate demand estimation results in inefficiencies across procurement, staffing, and storage.

Several factors influence food wastage, including fluctuating student attendance, irregular meal timings, special academic events, seasonal variations, and procurement practices. Traditional intuition-based planning is insufficient to capture these complex interactions. Consequently, a data-driven approach offers a systematic way to quantify patterns, identify inefficiencies, and propose targeted interventions.

This project adopts a Data Warehousing and Business Intelligence (DWBI) framework to analyze historical mess expenditure data as a proxy for consumption behavior. While direct food item-level wastage measurements are unavailable, financial transactions provide a reliable, consistently recorded dataset that can be leveraged to infer high-risk wastage periods and structural inefficiencies in procurement and vendor dependency. Raw data can be found here : http://mess.iisertvm.ac.in/expense.php.

Chapter 2

Data Sources and Collection Process

Data Sources

Although multiple mess units operate on campus—namely CDH-1, CDH-2, and the Café—all financial records are maintained centrally. Therefore, the project utilizes a single consolidated data source with multiple operational units embedded within it. The data is entirely digital and maintained in spreadsheet format. Each record corresponds to a vendor payment entry and contains attributes such as vendor name, bill information, payment date, amount paid, and administrative metadata. No handwritten or scanned documents were used. The data is primarily textual (vendor names, bill descriptors) and numerical (payment amounts), with date fields embedded both explicitly and implicitly within text columns. Payment data is recorded by mess administrative staff after receiving vendor bills. Bills are typically aggregated and paid periodically rather than daily, leading to a temporal mismatch between food consumption and recorded payment dates. Data is collected continuously and compiled into monthly spreadsheets. On average, each month contains between 250–400 transaction records. For scalability, the data can be extended to multiple hostels or campuses by introducing additional mess unit identifiers. A centralized data warehouse architecture was therefore adopted to support long-term storage, multi-month analysis, and future expansion.

Data Volume and Scope

The analysis covers the Aug-Dec 2025 academic semester. The dataset contains approximately 1,829 transaction entries recorded over the period. The number of raw transactions per month varies significantly, with peaks during semester start and mid-semester periods, and low activity during examination periods and breaks.

Transactions are classified by mess unit (CDH-1, CDH-2, Café, and UNKNOWN for miscellaneous or ambiguous entries). Each transaction includes vendor information, payment date, bill description, and monetary amount. While bill descriptors vary widely, they generally indicate the type of procurement (e.g., vegetables, provisions, milk, kitchen supplies).

Data Format and Initial Cleaning

The original dataset was maintained as monthly Excel files. Each month's file contained columns for bill number, bill date, vendor name, bill description, payment date, amount paid, and occasionally, administrative notes.

Inconsistencies in data entry required substantial preprocessing:

- Multiple vendor name spellings (e.g., "Rahul Stores," "Rahul Store," "Rahul's Store") required manual standardization.
- Payment dates were inconsistently formatted (DD-MM-YYYY, DD/MM/YY, textual dates like "15th August 2025").
- Bill descriptions varied from highly specific (e.g., "50 kg tomatoes") to vague (e.g., "provisions").
- Some records lacked vendor names or had placeholder values like "Cash Voucher" or "Miscellaneous."

To address these issues, vendor names were harmonized using a lookup table created through manual inspection and fuzzy matching. Dates were standardized to ISO 8601 format (YYYY-MM-DD) using automated parsing scripts with fallback manual validation. Records with missing or ambiguous vendor information were tagged under a special category called "UNKNOWN" to preserve data completeness while flagging anomalies for future resolution.

Ethical and Privacy Considerations

The data used in this study is derived from institutional financial records that are semi-public in nature. The Student Cooperative Mess maintains transparency in accounting by publishing monthly expense summaries. However, vendor names and transaction amounts were retained as-is because they represent institutional procurement behavior, not personal consumer data.

No personally identifiable information (PII) about students, staff, or individuals was collected or analyzed. The dataset contains only vendor names, institutional payment dates, and transaction amounts. All analysis was performed on aggregated data at the vendor, mess unit, and temporal levels. Individual transaction records were not disclosed beyond the scope of this project.

Absence of Item-Level Food Data

The data set does not contain information on items such as vegetables, grains, or meat quantities. This limitation is explicitly acknowledged. Since the objective is to identify systemic inefficiencies and high-risk periods rather than precise calorific wastage, expenditure-level analysis remains meaningful and operationally actionable. This rationale is highlighted to avoid misinterpretation of the scope of the project.

Vendor	Date	Bill Number and Date	Amount	Sheet Number	Status	Invigilator
Milan Agencies-Milk-CDH-1	01-01-2026	6425, 01-01-2026	6952.4	2026-01-01-4481	Processed	Secretary DAC
Cash Voucher CDH-1	01-01-2026	5657, 01-01-2026	240	2026-01-01-4506	Processed	Secretary DAC
Magom Rice Flowermill CDH-2	16-01-2026	303, 15-01-2026	900	2026-01-16-4511	Processed	Secretary DAC
Magom Rice Flowermill CDH-2	16-01-2026	97, 08-01-2026	780	2026-01-16-4511	Processed	Secretary DAC
Magom Rice Flowermill CDH-2	16-01-2026	92, 04-01-2026	1160	2026-01-16-4511	Processed	Secretary DAC
Magom Rice Flowermill CDH-2	16-01-2026	98, 11-01-2026	1400	2026-01-16-4511	Processed	Secretary DAC
Ponnus Papadam CDH-2	16-01-2026	31, 15-01-2026	12750	2026-01-16-4512	Processed	Secretary DAC
Ponnus Papadam CDH-2	16-01-2026	28, 09-01-2026	13600	2026-01-16-4512	Processed	Secretary DAC
Samudra Logistics CDH-2 cafe	16-01-2026	4999, 05-01-2026	12528	2026-01-16-4513	Processed	Secretary DAC
Thiruvathira Stores CDH-2CAFE	16-01-2026	117, 16-01-2026	9240	2026-01-16-4514	Processed	Secretary DAC
Priya Vegitables Chalai CDH-2	16-01-2026	738, 16-01-2026	65981	2026-01-16-4515	Processed	Secretary DAC
Priya Vegitables Chalai CDH-2	16-01-2026	729, 13-01-2026	60773	2026-01-16-4515	Processed	Secretary DAC
Priya Vegitables Chalai CDH-2	16-01-2026	702, 06-01-2026	58590	2026-01-16-4515	Processed	Secretary DAC
Priya Vegitables Chalai CDH-2	16-01-2026	720, 09-01-2026	103967	2026-01-16-4515	Processed	Secretary DAC
priya Vegitables Chalai CAFE-2	16-01-2026	739, 16-01-2026	8778	2026-01-16-4516	Processed	Secretary DAC
priya Vegitables Chalai CAFE-2	16-01-2026	730, 13-01-2026	4695	2026-01-16-4516	Processed	Secretary DAC
priya Vegitables Chalai CAFE-2	16-01-2026	703, 06-01-2026	11808	2026-01-16-4516	Processed	Secretary DAC
priya Vegitables Chalai CAFE-2	16-01-2026	721, 09-01-2026	6219	2026-01-16-4516	Processed	Secretary DAC
Star Bakery /cafe-cdh-2	16-01-2026	1920, 09-01-2026	17069	2026-01-16-4517	Processed	Secretary DAC
Star Bakery /cafe-cdh-2	16-01-2026	1919, 04-01-2026	14659	2026-01-16-4517	Processed	Secretary DAC
Star Bakery CDH-2	16-01-2026	1916, 01-01-2026	23100	2026-01-16-4518	Processed	Secretary DAC
Rahul Stores cdh-2 cafe	16-01-2026	1550, 15-01-2026	19640	2026-01-16-4519	Processed	Secretary DAC
Rahul Stores cdh-2 cafe	16-01-2026	1492, 02-01-2026	56924	2026-01-16-4519	Processed	Secretary DAC
Rahul Stores cdh-2 cafe	16-01-2026	1493, 02-01-2026	74429	2026-01-16-4519	Processed	Secretary DAC
Fathima Chicken Vithura CDH-2	16-01-2026	41, 15-01-2026	53668	2026-01-16-4520	Processed	Secretary DAC
Fathima Chicken Vithura CDH-2	16-01-2026	40, 12-01-2026	53792	2026-01-16-4520	Processed	Secretary DAC
Fathima Chicken Vithura CDH-2	16-01-2026	39, 08-01-2026	52515	2026-01-16-4520	Processed	Secretary DAC

Figure 2.1: Overview of the raw dataset. While the data are generally well maintained, several inconsistencies are evident(e.g., CDH-2 vs. cdh-2, CDH-1 vs. cdh-1). To address these issues, a data-cleaning step was applied in which all entries were stripped of leading and trailing whitespace and converted to uppercase using Python, ensuring consistent naming across the dataset.The preprocessing strategy focused on standardization rather than imputation. Records missing critical fields such as payment amount or date were excluded, as imputing financial transactions could introduce artificial bias. Vendor and mess unit names were normalized using rule-based mappings derived from observed variations.

Chapter 3

From Data Warehousing to Knowledge Discovery: Pipeline, Preprocessing, and System Constraints

The initial stage of the workflow involved data cleaning, where raw expenditure records extracted from a student cooperative mess portal contained inconsistencies such as missing payment dates, non-standard bill date formats, and textual vendor identifiers. These inconsistencies were resolved through systematic preprocessing using SQL, including date correction, normalization of vendor names, and removal of structurally invalid entries. This step ensured that downstream analytical operations were performed on reliable and consistent data.

Following data cleaning, data integration was performed by consolidating multiple monthly datasets (September to December 2025) into a unified relational schema. Although the source data originated from a single operational system, integration was required across time periods and logical entities such as vendors, mess units, and dates. This integration was achieved through a dimensional data warehouse design, consisting of fact and dimension tables, enabling consistent analysis across months.

Data selection was carried out using structured SQL queries to extract only analytically relevant attributes, such as payment date, vendor, mess unit, and transaction amount. These queries filtered the operational data into a subject-oriented format suitable for analysis, ensuring that irrelevant transactional metadata did not interfere with analytical clarity.

The transformation phase involved aggregating transactional-level data into daily totals, mess-group averages, and vendor-level summaries. Derived attributes such as estimated food waste were calculated using domain-informed proportional assumptions that link the magnitude of expenditure to the volume of waste. These transformations converted raw transactional data into higher-level analytical indicators suitable for trend detection and comparative analysis.

Data mining and pattern extraction were performed primarily through temporal trend analysis and statistical distribution analysis. Monthly expenditure patterns revealed strong correlations with academic calendar events such as mid- and end-session examinations, science festivals, and campus-wide events. Additionally, Benford's Law was applied at the vendor level to evaluate the naturalness of first-digit distributions in expenditure amounts, serving as a preliminary fraud detection heuristic within a business intelligence context.

Pattern evaluation was conducted by comparing observed trends with known institutional schedules and expected consumption behaviors. Peaks in expenditure and wastage coincided with increased campus occupancy, validating the plausibility of detected patterns. Deviations from expected statistical distributions were not interpreted as definitive fraud indicators but were flagged as candidates for deeper administrative review.

Finally, knowledge presentation was achieved through structured tabular summaries, time-series plots, bar charts, and comparative visualizations. These visual representations transformed complex numerical patterns into interpretable insights, enabling stakeholders to understand consumption dynamics, wastage behavior, and expenditure anomalies. Collectively, this pipeline demonstrates the practical application of data warehousing and data mining principles to a real-world institutional food systems problem.

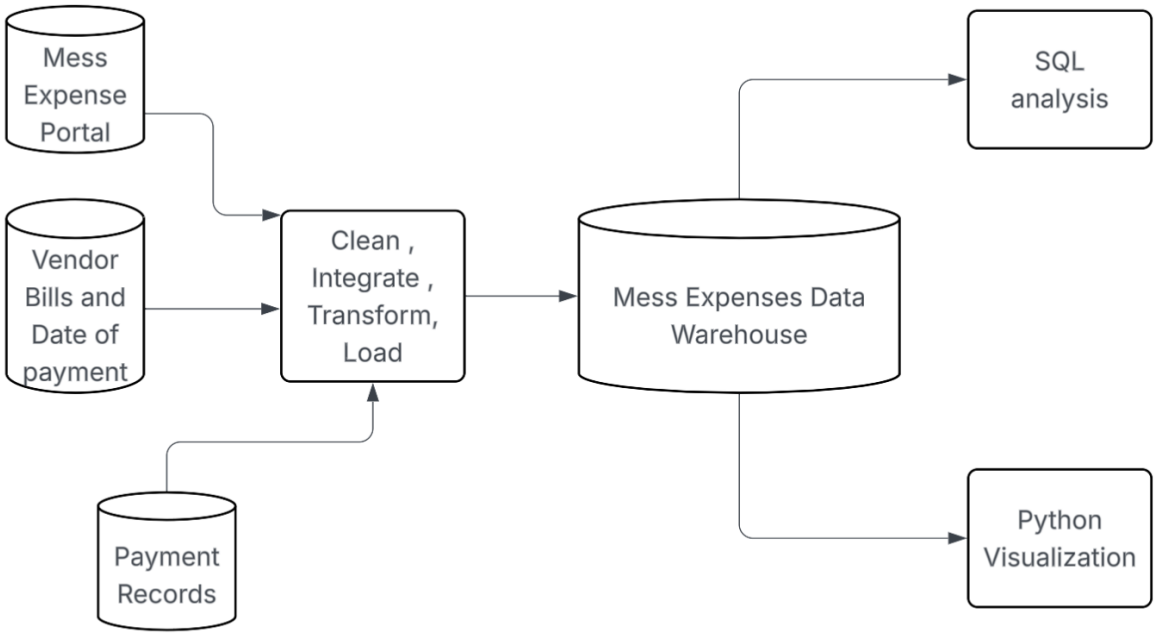


Figure 3.1: OLTP diagram of the database

Design and Structure of the Mess Expense Data Warehouse and OLAP

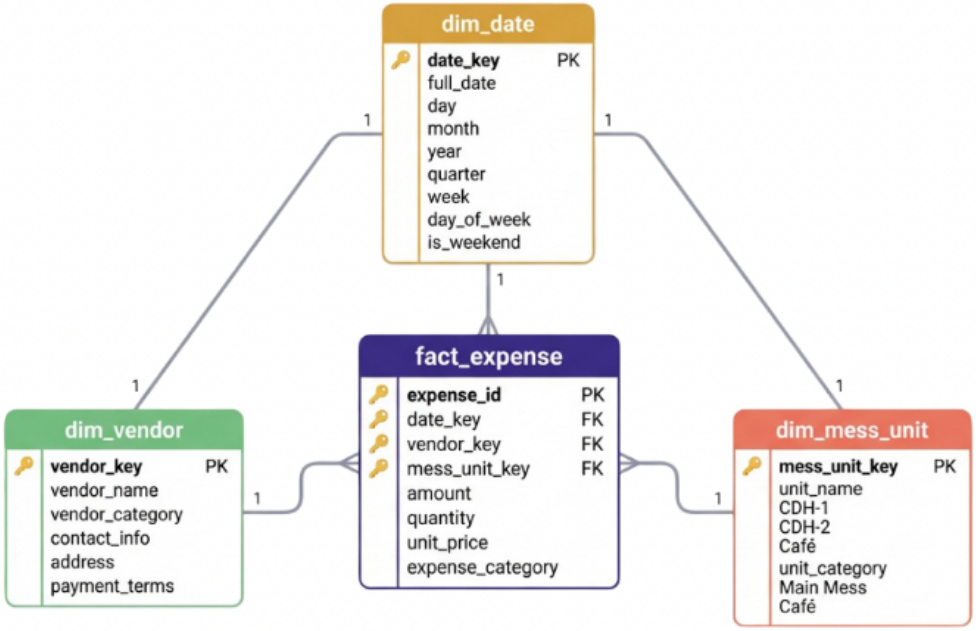


Figure 3.2: Star schema diagram showing fact table and dimensions tables

To enable structured analysis across multiple months and mess units, a centralized data warehouse was designed using a star schema architecture. The warehouse integrates raw transaction data into a consistent analytical format.

The warehouse consists of a central fact table capturing financial transactions and supporting dimension tables for dates, vendors, and mess units. Raw and staging tables are retained to preserve lineage and enable auditing.

ETL Process

Raw CSV files are first loaded into staging tables, where minimal assumptions are made about data quality. Dimensional tables are populated using distinct cleaned values, and validated records are then inserted into the fact table through controlled joins. This layered approach separates raw ingestion from analytical readiness.

Qualitative ETL Outcome

Table 3.2: Representative Records from August 2025 Staging Table

ID	Vendor	Cat	Unit	Date	Amount
19	Amma vegetables	1	CDH-1	08-08	100291
47	Rahul stores	2	CDH-2	08-06	149250
54	Priya Vegitables	2	CDH-2	08-05	105648
141	Amma vegetables	1	CDH-1	08-16	101000
137	Rahul Stores	1	CDH-1	08-11	184270

This warehouse structure enables efficient OLAP-style querying and forms the backbone for all analytical chapters that follow. The staging table clearly exhibits characteristics of raw operational data. Vendor names appear in multiple textual variants, categorical fields such as `vendor_category` contain heterogeneous values (numeric codes, text labels, and vendor-specific tags), and several attributes such as bill number and bill date are missing. These inconsistencies motivated the subsequent transformation phase.

The `vendor_category` field reflects raw classifications recorded at source and was not standardized prior to ingestion. Numeric values such as `1` and `2` broadly correspond to general provisions and perishable items respectively, while textual values such as `Milk` represent dairy suppliers. Café-related entries appear under multiple free-text labels. Since vendor categorization was not required for the analytical objectives of this study, these values were retained in their original form and excluded from downstream aggregation logic. During the transformation phase, vendor names were standardized using rule-based normalization to resolve spelling inconsistencies. Mess units were mapped into a controlled dimension consisting of CDH-1, CDH-2, CAFE, and UNKNOWN categories. Records that could not be reliably assigned to a mess unit were retained but excluded from comparative analyses to avoid classification bias. Missing bill-related fields were preserved as NULL values, as imputing such fields could introduce artificial structure not supported by the source data.

Limitations and Data Quality Considerations

While the data warehouse and analytical framework developed in this study enable comprehensive analysis of mess expenditure patterns, it is important to explicitly acknowledge certain limitations and data quality considerations inherent to the available data. Addressing these aspects strengthens the credibility of the analysis and ensures that interpretations are made within an appropriate methodological context.

Payment Date versus Consumption Date Mismatch

The primary limitation arises from the fact that the available transactional data records *payment dates* rather than actual *food consumption dates*. In the operational workflow of the student mess, vendor payments are often settled in batches, with multiple bills corresponding to several days or weeks of consumption being cleared on a single payment date. Consequently, spikes observed in daily expenditure values do not necessarily indicate unusually high food consumption on those specific dates, but rather reflect the settlement of accumulated liabilities.

All temporal analyses in this study therefore interpret expenditure patterns as financial settlement behavior rather than direct consumption behavior. This distinction is consistently maintained throughout the analysis to avoid misrepresentation of daily food demand.

Aggregation Bias in Daily Expenditure Analysis

The aggregation of multiple vendor payments into a single daily total introduces aggregation bias, particularly when comparing daily expenditure values across the semester. Days with exceptionally high totals may represent delayed or bulk payments, while days with low or zero expenditure may still correspond to normal levels of food consumption that were not immediately settled financially.

Despite this limitation, aggregation remains appropriate for business intelligence purposes, as the objective of the study is to analyze financial flows, budgeting behavior, and expenditure risk patterns rather than precise per-day consumption quantities.

Proxy-Based Wastage Estimation Assumptions

Food wastage in this study is estimated using a proxy model based on total expenditure values, due to the absence of direct wastage measurement data. This approach assumes a proportional relationship between expenditure and potential wastage, enabling relative comparison across dates, mess units, and academic periods.

It is important to note that the estimated wastage values should be interpreted as *risk indicators* rather than absolute measurements. The model is designed to identify periods of unusually high financial outflow that may warrant closer operational scrutiny, rather than to quantify exact food waste in kilograms. The estimated wastage for a given day d is computed as:

$$W(d) = \frac{E(d)}{\bar{E}_{\text{month}}} \times k \tag{3.1}$$

where $E(d)$ is the total expense on day d, $\bar{E}_{\text{month}}$ is the monthly average daily expense, and $k = 100\,\text{kg}$ is a domain-informed reference constant representing the estimated daily wastage at average expenditure levels.

Lack of Item-Level Transactional Data

The dataset does not contain item-level details such as quantities of specific food items procured, menu composition, or per-meal consumption statistics. As a result, the analysis operates at the vendor and mess-unit level, limiting the ability to attribute expenditure changes to specific ingredients or menu decisions.

While item-level data would enable more granular operational optimization, the current vendor-level aggregation remains sufficient for strategic financial analysis, vendor dependency assessment, and structural network analysis of procurement relationships.

Overall Impact on Analytical Validity

Despite these limitations, the data quality is adequate for the stated objectives of the project, which focus on financial behavior, expenditure trends, structural dependencies, and anomaly detection rather than precise consumption accounting. The consistent application of assumptions across all months and mess units ensures internal comparability, making the derived insights reliable within the defined analytical scope.

Explicit recognition of these limitations enhances the transparency and robustness of the study and provides a clear foundation for future extensions involving higher-resolution operational data.

Chapter 4

Month-wise Analysis

August 2025 - The Semester Begins

Each record contains the vendor name, bill information, and the total amount paid. Since the dataset lacked explicit mess unit identifiers and a dedicated date column, preprocessing was required before analysis. The data was ingested into a MySQL staging table, where expense dates were extracted from bill information using SQL string operations and converted into a standard `DATE` format. Mess units were inferred using rule-based classification on vendor names, mapping records to CDH-1, CDH-2, or Café. Transactions that could not be reliably classified, such as cash vouchers and shared administrative expenses, were labeled as `UNKNOWN` and excluded from comparative mess-level analysis.

A star-schema data warehouse was constructed to support analytical queries. The warehouse consists of a central fact table (`fact_expense`) containing transaction-level expenditure data, linked to dimension tables for date (`dim_date`), vendor (`dim_vendor`), and mess unit (`dim_mess_unit`). SQL queries were used to aggregate daily expenses, which were then analyzed using Python for trend identification and visualization.

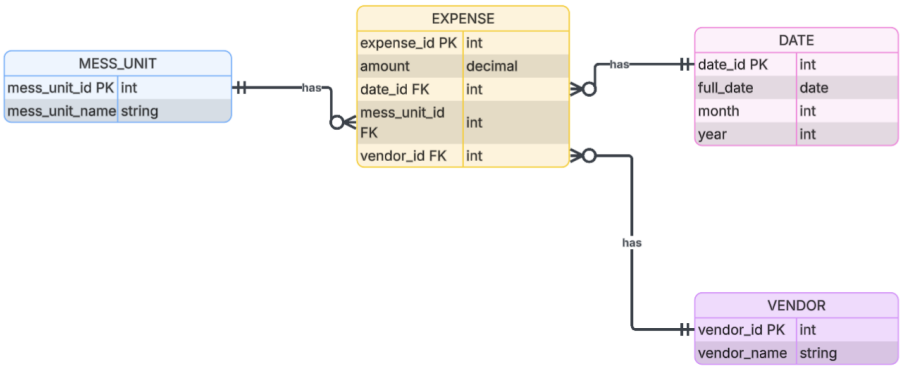


Figure 4.1: Entity-Relationship Diagram (ERD) for the database created.

Table 4.1: Daily Total Mess Expense for August 2025

Date	Total Expense (Rs.)
2025-07-29	4,966.00
2025-08-01	250,787.80
2025-08-02	92,405.00
2025-08-03	31,216.10
2025-08-04	173,552.00
2025-08-05	438,681.70
2025-08-06	504,113.50
2025-08-07	252,638.90
2025-08-08	299,550.50
2025-08-09	96,592.50
2025-08-10	42,104.00
2025-08-11	493,431.80
2025-08-12	390,719.20
2025-08-13	35,992.30
2025-08-14	201,741.00
2025-08-15	219,135.55
2025-08-16	213,184.40
2025-08-17	39,432.50
2025-08-18	73,871.30
2025-08-19	287,289.60
2025-08-20	391,160.00
2025-08-21	116,897.30
2025-08-22	280,287.20
2025-08-23	51,847.50

The average daily expense for the month was calculated to be **Rs. 2,07,566.57**. Days with expenses exceeding this value were identified as *high-wastage risk days*, under the assumption that higher expenditure correlates with increased food preparation and procurement, thereby increasing the likelihood of food wastage.

Table 4.2: High-Wastage Risk Days

Date	Total Expense (Rs.)
2025-08-01	250,787.80
2025-08-05	438,681.70
2025-08-06	504,113.50
2025-08-07	252,638.90
2025-08-08	299,550.50
2025-08-11	493,431.80
2025-08-12	390,719.20
2025-08-15	219,135.55
2025-08-16	213,184.40
2025-08-19	287,289.60
2025-08-20	391,160.00
2025-08-22	280,287.20

Table 4.3: Estimated Food Wastage

Date	Expense (Rs.)	Wastage (kg)
2025-07-29	4,966.00	2.39
2025-08-01	250,787.80	120.82
2025-08-02	92,405.00	44.52
2025-08-03	31,216.10	15.04
2025-08-04	173,552.00	83.61
2025-08-05	438,681.70	211.35
2025-08-06	504,113.50	242.87
2025-08-07	252,638.90	121.71
2025-08-08	299,550.50	144.32
2025-08-09	96,592.50	46.54
2025-08-10	42,104.00	20.28
2025-08-11	493,431.80	237.72
2025-08-12	390,719.20	188.24
2025-08-13	35,992.30	17.34
2025-08-14	201,741.00	97.19
2025-08-15	219,135.55	105.57
2025-08-16	213,184.40	102.71
2025-08-17	39,432.50	19.00
2025-08-18	73,871.30	35.59
2025-08-19	287,289.60	138.41
2025-08-20	391,160.00	188.45
2025-08-21	116,897.30	56.32
2025-08-22	280,287.20	135.03
2025-08-23	51,847.50	24.98

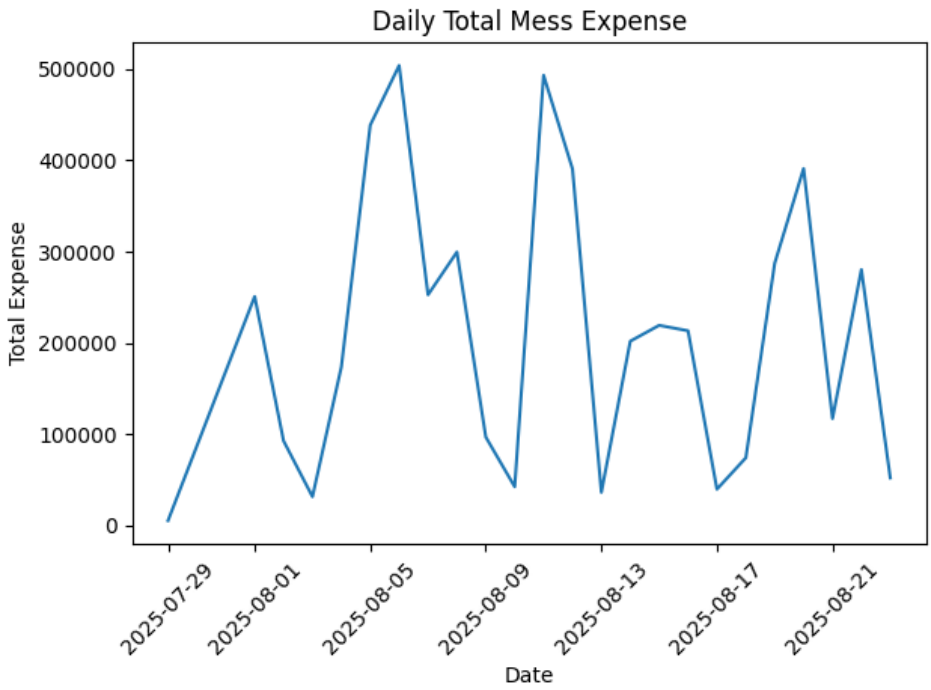
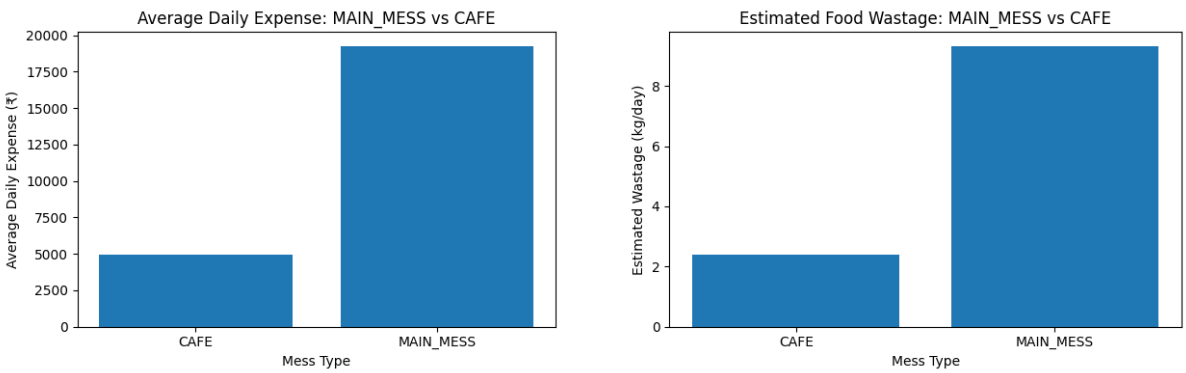


Figure 4.2: Daily Total Mess Expense Trend for August 2025. A pronounced spike in total expenditure is observed during the first two weeks of the semester, corresponding to the post–summer vacation influx of students.



(a) Average Daily Expense

(b) Estimated Food Wastage

Figure 4.3: Comparison Between Main Mess and Café

For comparative performance analysis, CDH-1 and CDH-2 were combined into a single Main Mess category, while Café operations were analyzed separately. Transactions labeled as `UNKNOWN` were excluded. The comparison indicates that Café operations exhibit higher relative expenditure volatility and higher estimated wastage per unit expense compared to the Main Mess, reflecting irregular demand patterns and reduced predictability.

September 2025 – The Month of Mid-Semester Examinations

September 2025 corresponds to a critical academic phase due to the commencement of mid-semester examinations around the third week of the month. This period is typically associated with changes in student activity patterns, extended study hours, increased café usage, and altered food consumption behavior. The following table presents the daily total mess expenditure and estimated wastage for each payment date in September 2025.

Table 4.4: Daily Expense with Estimated Food Wastage for September 2025

Date	Expense (Rs.)	Wastage (kg)
2025-09-01	202596.30	71.90
2025-09-02	1120119.60	397.50
2025-09-03	605340.60	214.82
2025-09-04	350151.90	124.26
2025-09-05	38046.00	13.50
2025-09-06	57508.50	20.41
2025-09-07	17056.50	6.05
2025-09-08	160001.40	56.78
2025-09-09	436371.90	154.86
2025-09-10	158896.50	56.39
2025-09-11	593793.40	210.72
2025-09-12	282928.90	100.40
2025-09-13	102785.70	36.48
2025-09-14	62354.50	22.13
2025-09-15	201242.30	71.42
2025-09-16	402985.05	143.01
2025-09-17	312381.60	110.86
2025-09-18	486320.80	172.58
2025-09-19	696649.90	247.22
2025-09-20	118013.50	41.88
2025-09-21	53465.00	18.97
2025-09-22	150539.40	53.42
2025-09-23	359252.30	127.49
2025-09-24	225701.90	80.10
2025-09-25	131142.60	46.54
2025-09-26	471503.20	167.32
2025-09-27	79349.40	28.16
2025-09-28	13656.50	4.85

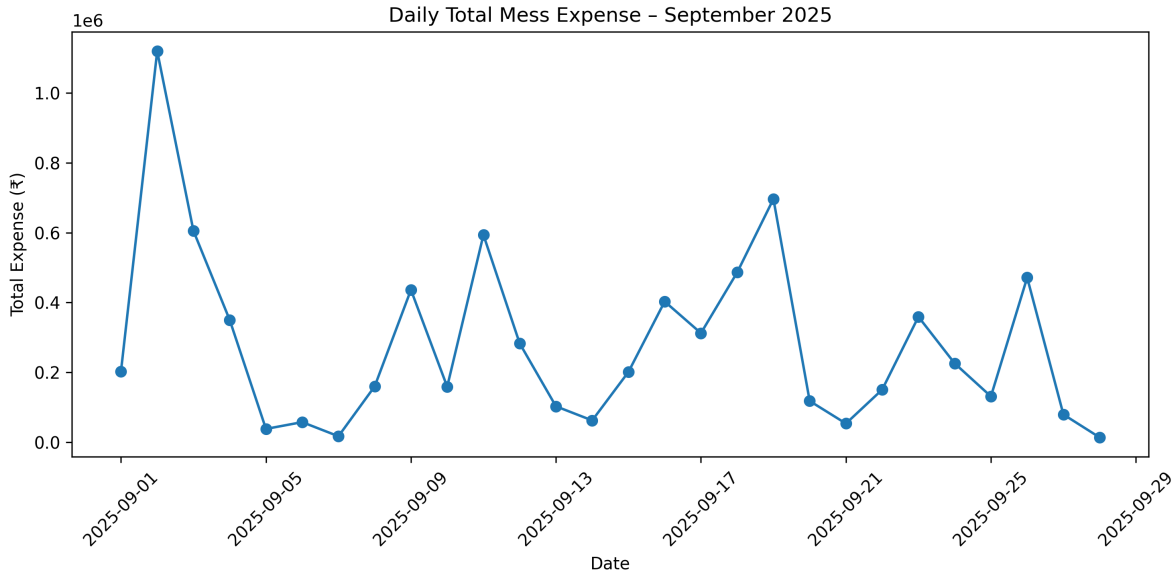


Figure 4.4: The spikes observed on 1st September and around mid-September can be explained by operational and behavioral factors. The increase on 1st September corresponds to the beginning of a new month, during which bulk procurement is typically undertaken to replenish inventory. The subsequent spike observed around mid-September aligns with the period preceding mid-semester examinations (commencing around 20th September). During this phase, students tend to shift their food consumption from the central mess to campus cafés due to extended study hours, leading to demand volatility and procurement–consumption mismatch within the mess.

The average daily expenditure for September 2025, computed directly from the daily totals listed above, was **Rs. 281,791.26**. Days on which the total mess expenditure exceeded this average were flagged as high-wastage-risk days, as elevated spending often correlates with over-procurement and increased food wastage. The dates identified through this criterion are presented alongside. Since direct food waste measurements were not available, waste was estimated using a proportional proxy model scaled against the monthly average daily expense.

Table 4.5: High Wastage Days for September 2025

Date	Total Expense (Rs.)
2025-09-02	1,120,119.60
2025-09-03	605,340.60
2025-09-04	350,151.90
2025-09-09	436,371.90
2025-09-11	593,793.40
2025-09-12	282,928.90
2025-09-16	402,985.05
2025-09-17	312,381.60
2025-09-18	486,320.80
2025-09-19	696,649.90
2025-09-23	359,252.30
2025-09-26	471,503.20

Overall, the observed data trends align with our expectations. During the mid-semester examination period, there was an increase in food demand and irregular consumption patterns, consistent with anticipated student behavior. Figure 4.4 further illustrates the expected decrease in total mess expenditure following the conclusion of exams on 27th September, as many students returned home. A total of 390 records were analyzed to assess the mess expenses for September.

October 2025 – The Month of Science and Sports Fests

October 2025 marks a distinct operational phase in the mess ecosystem. Unlike September, which was dominated by academic pressures due to mid-semester examinations, October witnessed a sharp resurgence in campus activity driven by the return of students after semester breaks and the hosting of major academic and cultural events. From around 5th October onwards, student footfall increased substantially, which is clearly reflected in the expenditure and estimated food wastage trends observed during the month.

The average daily mess expenditure for October 2025 was Rs. 265,018.59. Several days exhibited significantly higher-than-average expenses, particularly 2nd, 4th, 7th, 10th, 13th, 14th, 15th, 16th, 17th, 20th, and 24th October. These spikes coincide with the timeline of major campus events. The dominance of the Main Mess in both expenditure and estimated food wastage remains consistent; however, the Café exhibits a noticeable rise in activity during this period. This aligns strongly with the timing of two major campus events — **ANVESHA**, one of the largest science festivals, and **ITSAV**, a multi-sport cultural event.

Table 4.6: Daily Expense with Estimated Food Wastage for October 2025

Date	Expense (Rs.)	Wastage (kg)
2025-10-01	135524.60	51.14
2025-10-02	594604.60	224.36
2025-10-03	170413.70	64.30
2025-10-04	671767.00	253.48
2025-10-05	42931.00	16.20
2025-10-06	116067.40	43.80
2025-10-07	590569.05	222.84
2025-10-08	233508.90	88.11
2025-10-09	139453.50	52.62
2025-10-10	335743.50	126.69
2025-10-11	104014.30	39.25
2025-10-12	64322.70	24.27
2025-10-13	327927.70	123.74
2025-10-14	395754.80	149.33
2025-10-15	439174.40	165.71
2025-10-16	301777.90	113.87
2025-10-17	347779.60	131.23
2025-10-18	155347.00	58.62
2025-10-19	122469.00	46.21
2025-10-20	285274.90	107.64
2025-10-21	225528.00	85.10
2025-10-22	113787.50	42.94
2025-10-23	149616.50	56.46
2025-10-24	544229.60	205.36
2025-10-25	17877.60	6.75

Table 4.7: High Wastage Days for October 2025

Date	Expense (Rs.)
2025-10-02	594,604.60
2025-10-04	671,767.00
2025-10-07	590,569.05
2025-10-10	335,743.50
2025-10-13	327,927.70
2025-10-14	395,754.80
2025-10-15	439,174.40
2025-10-16	301,777.90
2025-10-17	347,779.60
2025-10-20	285,274.90
2025-10-24	544,229.60

Table 4.8: Average Daily Expense and Estimated Wastage by Mess Group (October 2025)

Mess Group	Avg. Expense (Rs.)	Avg. Wastage (kg)
CAFÉ	11066.74	4.18
MAIN_MESS	23244.03	8.77

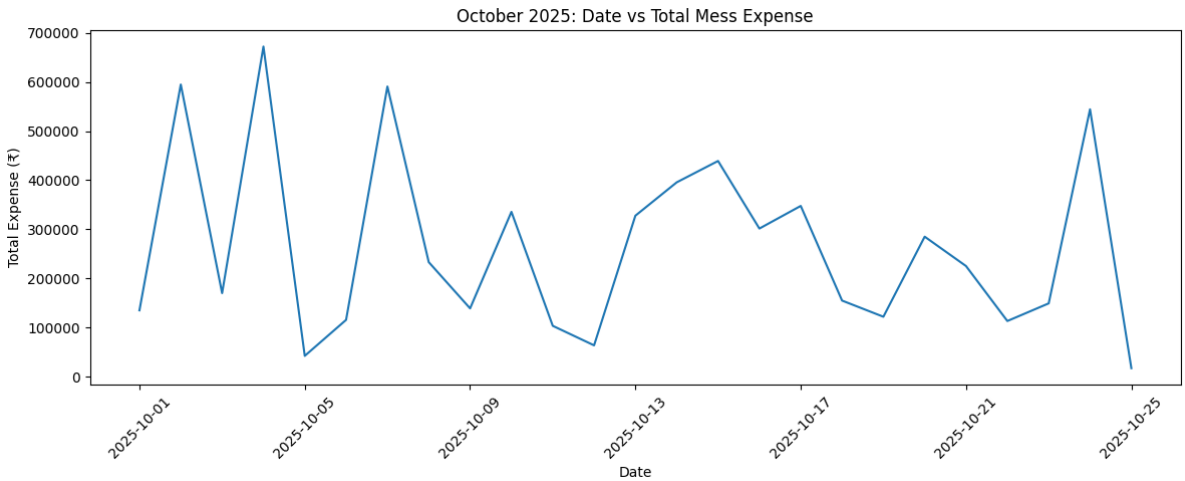


Figure 4.7: Date vs Total Mess Expenditure for October 2025

Overall, October 2025 represents a month where expenditure patterns deviate from routine academic cycles and are instead driven by institutional events and student congregation dynamics. While the observed trends may appear intuitive, their quantitative confirmation through data-driven analysis is essential for validating operational assumptions and forms the foundation for more advanced predictive and optimization models in subsequent phases of this project.

November 2025 – The Month of End Semester Examinations

November 2025 represents a crucial academic phase, marked by the commencement of end semester examinations from 19th November onward. This transition is clearly reflected in the mess expenditure patterns observed during the month. As students returned to campus and academic intensity increased, both food consumption and associated operational expenses exhibited noticeable fluctuations. The following analysis presents a detailed examination of daily expenditure trends, estimated food wastage, and mess-wise consumption behavior for November 2025.

Using the predefined wastage conversion factor, estimated food wastage was calculated for each day based on total daily expenditure. Table 4.9 presents the full dataset, including every recorded date, total expense, and corresponding estimated food wastage in kilograms. An aggregation by mess group reveals a clear difference between Café and Main Mess consumption patterns. The Main Mess consistently accounts for a higher average daily expense and estimated food wastage, reflecting its role as the primary dining facility during academically intensive periods such as end semester examinations. Overall, November 2025 demonstrates a strong linkage between academic schedules and food consumption behavior. The alignment of expenditure peaks with examination periods reinforces the importance of incorporating academic calendars into operational planning and wastage mitigation strategies for institutional food systems.

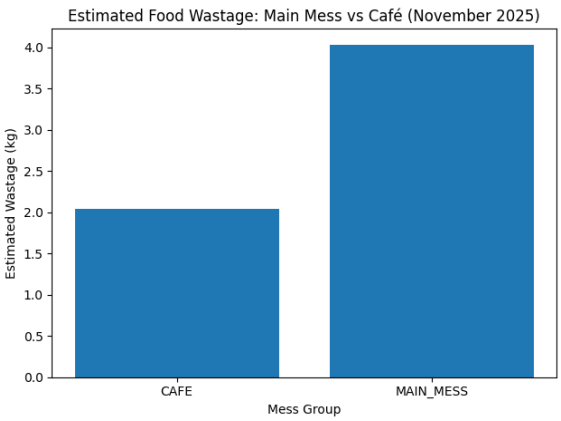


Figure 4.8: Estimated Food Wastage Comparison Between Main Mess and Cafe

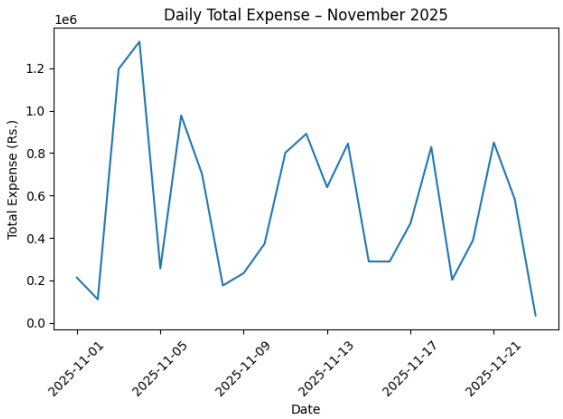


Figure 4.9: Daily Total Mess Expense for November 2025

Table 4.9: Daily Expense with Estimated Food Wastage (November 2025)

Date	Amount (Rs.)	Wastage (kg)
2025-11-01	212433.40	38.57
2025-11-02	110274.20	20.02
2025-11-03	1196157.20	217.19
2025-11-04	1324937.60	240.57
2025-11-05	255270.20	46.35
2025-11-06	977196.20	177.43
2025-11-07	700999.30	127.28
2025-11-08	175488.20	31.86
2025-11-09	234207.20	42.53
2025-11-10	372503.80	67.64
2025-11-11	801512.00	145.53
2025-11-12	890479.00	161.69
2025-11-13	638829.00	115.99
2025-11-14	844426.20	153.32
2025-11-15	288885.00	52.45
2025-11-16	288598.60	52.40
2025-11-17	468070.00	84.99
2025-11-18	828691.00	150.47
2025-11-19	202563.00	36.78
2025-11-20	388846.00	70.60
2025-11-21	849324.40	154.21
2025-11-22	583272.20	105.91
2025-11-23	34250.32	6.22

Table 4.10: High Wastage Days for November 2025

Date	Total Expense (Rs.)
2025-11-03	1196157.20
2025-11-04	1324937.60
2025-11-06	977196.20
2025-11-07	700999.30
2025-11-11	801512.00
2025-11-12	890479.00
2025-11-13	638829.00
2025-11-14	844426.20
2025-11-18	828691.00
2025-11-21	849324.40
2025-11-22	583272.20

December 2025 — End of Semester and Campus Departure

December 2025 represents the final operational phase of the student mess. Mess operations during this month were active only until 9th December, which was the last working day before the semester break. A pronounced decline in food expenditure is observed toward the end of the month, corresponding to the conclusion of end-semester examinations on 3rd December and the subsequent departure of students from campus.

The expense data for this month therefore captures a transition period, beginning with residual high demand immediately after examinations and ending with minimal activity as the campus

population rapidly decreased.

Table 4.11: Daily Expense with Estimated Food Wastage — December 2025

Date	Expense (Rs.)	Wastage (kg)
2025-12-01	783448.10	346.56
2025-12-02	228160.10	100.93
2025-12-03	318818.00	141.03
2025-12-04	279594.70	123.68
2025-12-05	130693.20	57.81
2025-12-06	41159.20	18.21
2025-12-07	54994.20	24.33
2025-12-08	182834.60	80.88
2025-12-09	14898.00	6.59

The average daily expenditure for the operational days in December was Rs.**226,066.68**. The highest expenditure occurred on 1st December, reflecting continued campus occupancy immediately following examinations, while a sharp decline is visible after 5th December as students began vacating the campus.

Table 4.12: High Wastage Days — December 2025

Date	Total Expense (Rs.)
2025-12-01	783,448.10
2025-12-02	228,160.10
2025-12-03	318,818.00
2025-12-04	279,594.70

Table 4.13: Average Daily Expense and Estimated Wastage by Mess Group — December 2025

Mess Group	Avg. Expense (Rs.)	Wastage (kg/day)
Café	13,990.68	6.19
Main Mess	18,401.02	8.14

These days fall immediately before and shortly after the end of examinations, a period characterized by uncertainty in student departure schedules and transitional consumption behavior.

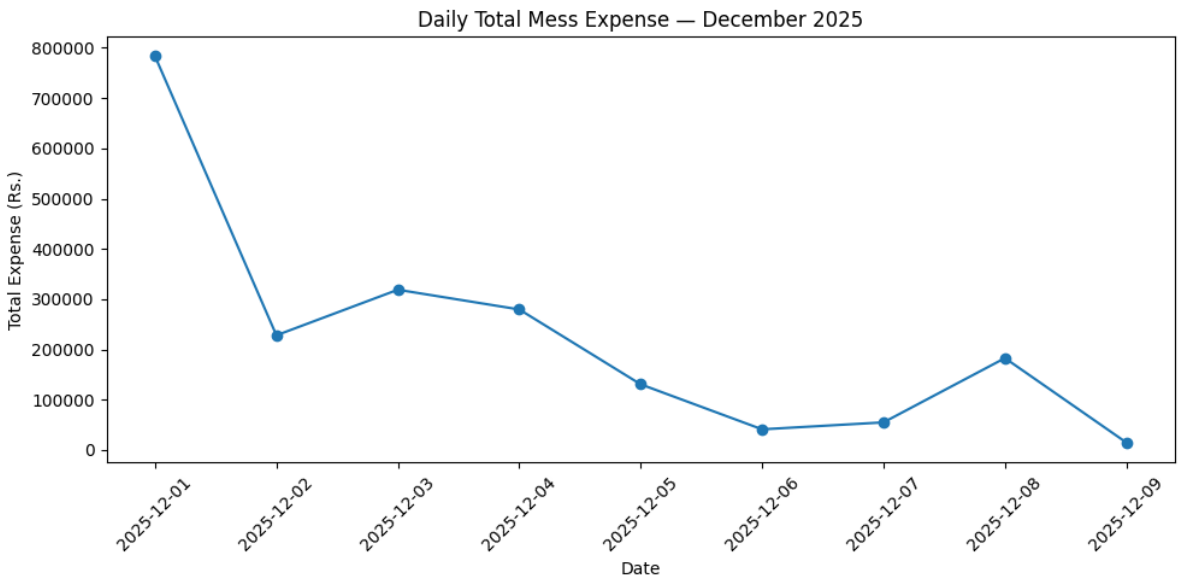


Figure 4.10: Daily Total Mess Expenditure — December 2025

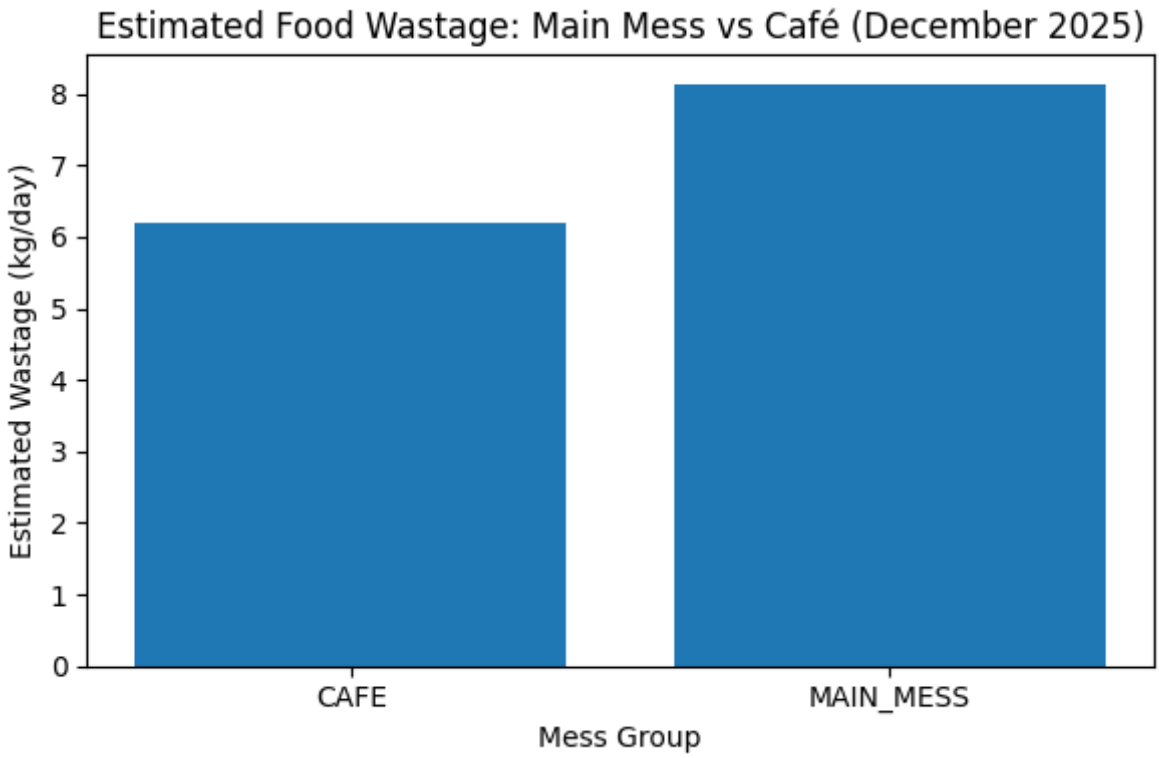


Figure 4.11: Estimated Food Wastage by Mess Group — December 2025

December exhibits a structurally different demand pattern compared to earlier months. Unlike spikes driven by academic events or festivals, expenditure dynamics here are dominated by campus attrition. The Main Mess shows higher average wastage than the Café, reflecting the difficulty of rapidly scaling down bulk meal preparation in response to sudden population

decline. Café operations, being more demand-driven and flexible, adapt more efficiently during such transition periods.

The December 2025 analysis highlights the importance of adaptive procurement strategies during semester transitions. The observed decline in expenditure and wastage after early December confirms that demand forecasting during exit phases is as critical as during peak academic activity. Incorporating academic calendar milestones into procurement planning can substantially reduce food wastage during semester-end periods.

Chapter 5

Benford's Law Analysis of Mess Expenses

While monthly trend analysis provides valuable insights into operational behavior, business intelligence systems are most effective when they enable deeper diagnostic and forensic analytics. To extend this study beyond descriptive reporting, Benford's Law was applied to vendor-level mess expense data for Aug-Dec 2025 as a preliminary fraud and anomaly detection technique.

Benford's Law, also known as the First-Digit Law, describes the empirical observation that in many naturally occurring numerical datasets, the leading digits are not uniformly distributed. Instead, smaller digits appear disproportionately often. The probability that a number has first digit $d \in \{1, 2, \ldots, 9\}$ is given by the logarithmic relationship:

$$P(d) = \log_{10}\left(1 + \frac{1}{d}\right)$$

This results in the digit 1 appearing as the leading digit approximately 30.1% of the time, while higher digits occur with progressively lower frequencies. This phenomenon has been extensively validated across financial, demographic, and economic datasets and is widely used in forensic accounting and audit analytics.

The applicability of Benford's Law relies on datasets that span multiple orders of magnitude and arise from natural economic processes rather than artificially assigned values. Vendor-level expense totals satisfy these criteria, as they are generated through repeated procurement transactions involving diverse suppliers and varying order sizes. Consequently, Benford's Law serves as a suitable screening tool for the mess expense data warehouse constructed in this project.

For this analysis, the leading digit of each individual transaction amount was extracted from the 1,829 records in the fact table. The observed frequency distribution was compared against the theoretical Benford distribution. Figure 5.1 presents this comparison.

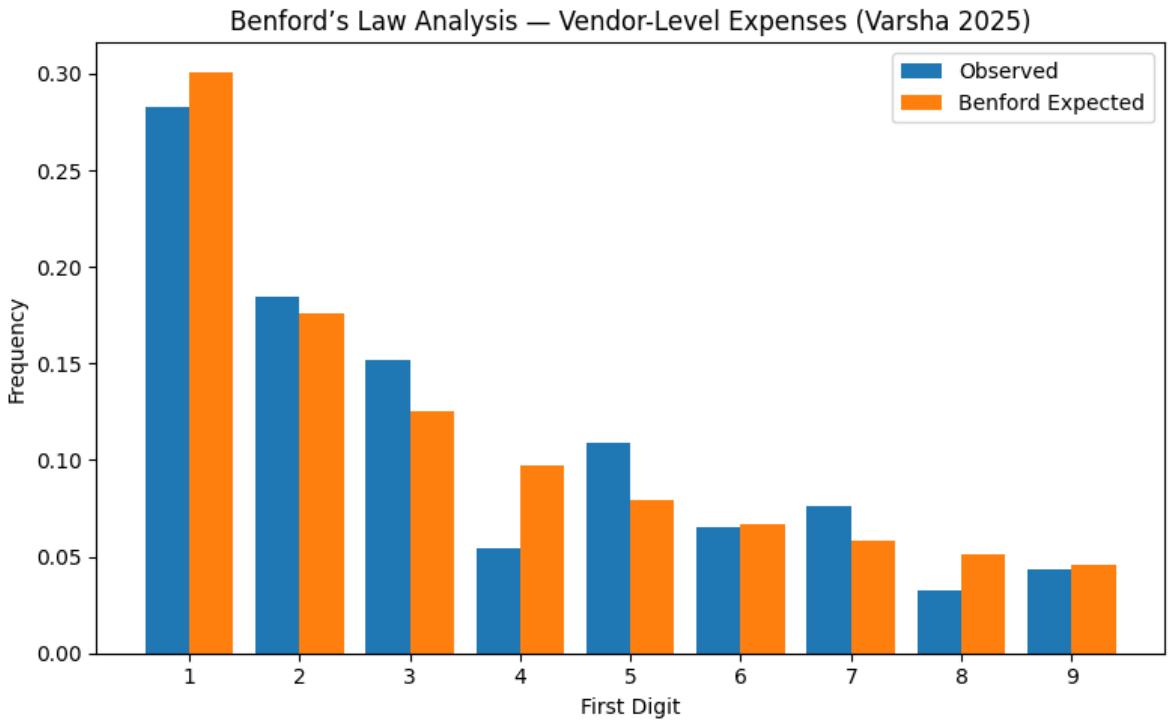


Figure 5.1: Benford's Law Analysis of Vendor-Level Mess Expenses

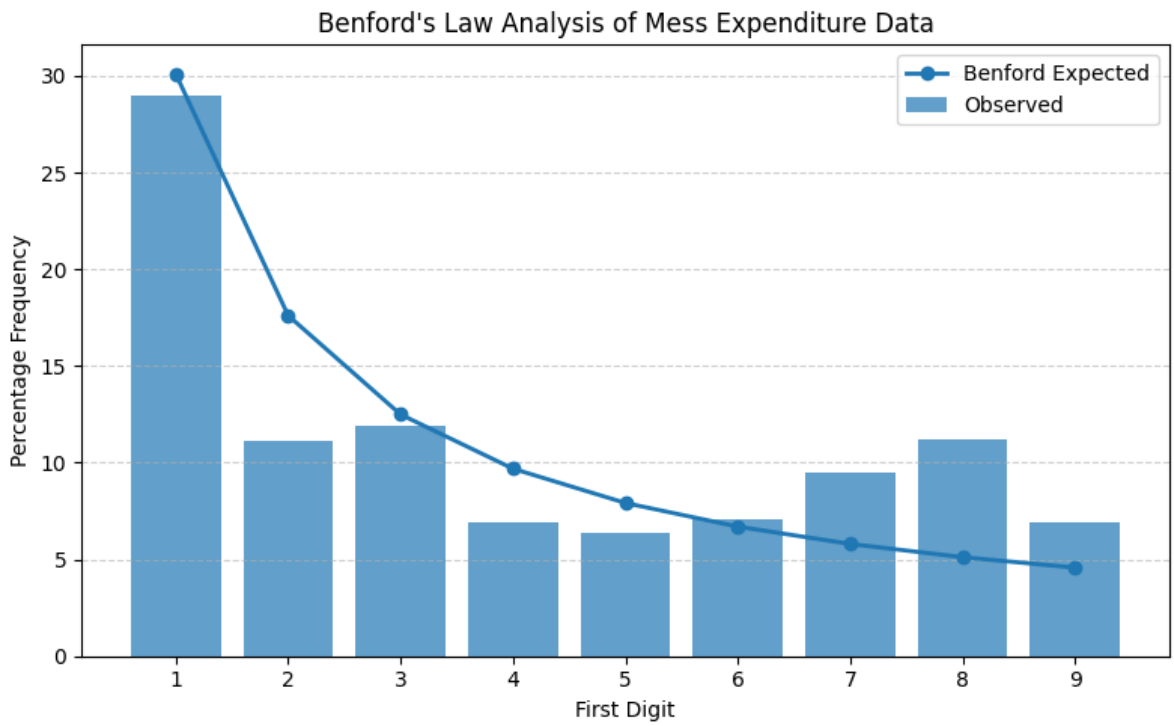


Figure 5.2: Observed vs Benford-Expected Leading Digit Distribution for Mess Expenditure Data

Methodology

The analytical workflow consisted of the following steps:

1. Extraction of vendor-level expenditure values from the data warehouse.
2. Removal of zero or null values to avoid undefined leading digits.
3. Extraction of the first non-zero digit from each transaction amount.
4. Computation of observed frequency counts for digits 1 through 9.
5. Calculation of expected frequencies using Benford's Law.
6. Statistical comparison using a chi-square goodness-of-fit test.

Statistical Validation: Chi-Square Goodness-of-Fit Test

To quantitatively assess the deviation, a chi-square goodness-of-fit test was performed at a 5% significance level.

The test statistic is defined as:

$$\chi^2 = \sum_{d=1}^{9} \frac{(O_d - E_d)^2}{E_d} \tag{5.1}$$

where:

1. O_d is the observed count for digit d
2. E_d is the expected count based on Benford's Law = Total transaction entries · P(d) = 1829 · P(d)

Test Results

- Chi-square statistic: 261.70
- Degrees of freedom: 8
- Critical value at $\alpha = 0.05$: 15.51
- P-value: < 0.001

Since the computed chi-square statistic greatly exceeds the critical value, the null hypothesis that the data follows Benford's Law is rejected at the 5% significance level.

Observed and Expected Frequencies

Table 5.1 presents the detailed comparison between observed and expected digit frequencies.

Table 5.1: Observed vs Expected Leading Digit Frequencies

Digit	Obs.	Exp.	Obs. %	Exp. %
1	530	550.58	28.98	30.10
2	203	322.07	11.10	17.61
3	218	228.51	11.92	12.49
4	127	177.25	6.94	9.69
5	117	144.82	6.40	7.92
6	130	122.45	7.11	6.69
7	173	106.07	9.46	5.80
8	205	93.56	11.21	5.12
9	126	83.69	6.89	4.58

Limitations

It is important to acknowledge that Benford's Law is sensitive to data generation mechanisms. Institutional procurement systems may violate some assumptions of randomness due to negotiated pricing, fixed-rate contracts, and batch payments. Therefore, Benford's Law should be interpreted as an exploratory diagnostic tool rather than definitive proof of irregularity.

Conclusion

The application of Benford's Law to vendor-level mess expenditure data provides an additional analytical dimension beyond descriptive statistics and trend analysis. By combining visual inspection with formal statistical testing, this study demonstrates how forensic data mining techniques can be integrated into data warehousing and business intelligence workflows to enhance financial transparency and audit readiness.

Chapter 6

Network Analysis of the Mess–Vendor Financial System

While earlier sections focused on temporal expenditure trends and aggregate financial behavior, the objective here is to uncover the *structural organization* of financial relationships between mess units and vendors. By modeling the system as a weighted graph, this section identifies dependency patterns, financial hubs, and naturally emerging communities within the mess procurement network.

Network Construction from the Data Warehouse

The mess–vendor financial network was constructed directly from the data warehouse by integrating information from the fact and dimension tables. Each node in the network represents either a mess entity (CDH-1, CDH-2, Café, and UNKNOWN) or an individual vendor. An undirected edge exists between a vendor and a mess unit if at least one financial transaction occurred between them. The weight assigned to each edge corresponds to the **total monetary value of transactions** between that vendor and mess unit over the academic year.

This modeling approach transforms transactional data into a relational structure, allowing financial dependencies to be analyzed as network properties rather than isolated records. The resulting network contains **96 nodes** and **114 edges**, capturing the complete vendor–mess interaction landscape without aggregation or pruning.

Global Network Characteristics

The global structure of the network exhibits a hub-and-spoke topology, where mess units act as central hubs connected to multiple vendors. These nodes serve as the primary financial anchors of the system, while several vendors remain sparsely connected, reflecting low-frequency or specialized procurement relationships. A notable feature of the network is the presence of

the *UNKNOWN* node, which aggregates expenses such as cash vouchers and administrative transactions. Its relatively high connectivity highlights the importance of improving expense classification and auditing mechanisms within the financial workflow.

Centrality and Financial Dependency Analysis

To quantify structural importance, degree centrality and weighted degree were computed for each node. Degree centrality measures the number of direct connections a node has, while weighted degree captures the total financial volume flowing through that node.

The mess units CDH-1 and CDH-2 emerge as the most dominant nodes, with weighted degrees of Rs.15.54 million and Rs 15.38 million respectively. Café also exhibits high degree centrality, reflecting consistent interactions with multiple vendors. Several vendors, including *Amrithamayi Charitable Society*, *Amma Vegetables*, and *Rahul Stores*, appear as significant financial dependency points due to sustained high-value transactions with specific mess units.

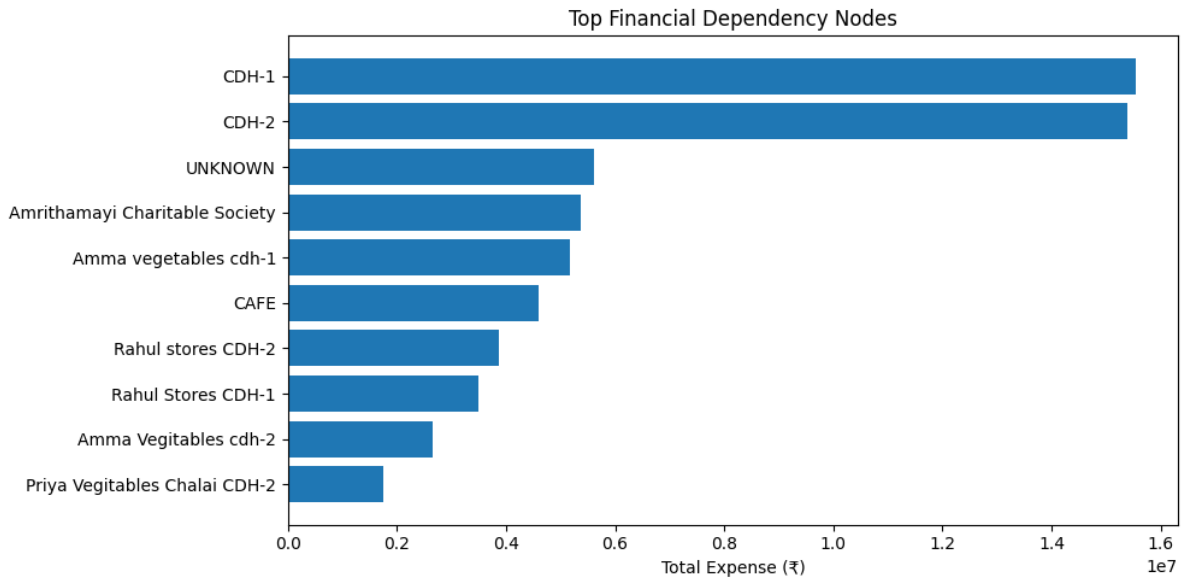


Figure 6.1: Node Importance Analysis

These high-weight vendors represent potential *risk concentration points*, where operational or pricing disruptions could disproportionately impact mess operations.

Community Detection and Modularity

To uncover latent organizational structure, community detection was performed using a modularity-maximization algorithm. Importantly, the number of communities was **not predefined**. Instead, the algorithm autonomously partitioned the network into **four communities**, achieving a modularity score of **0.650**. Such a high modularity value indicates strong separation between communities, meaning nodes are far more densely connected within communities than across them.

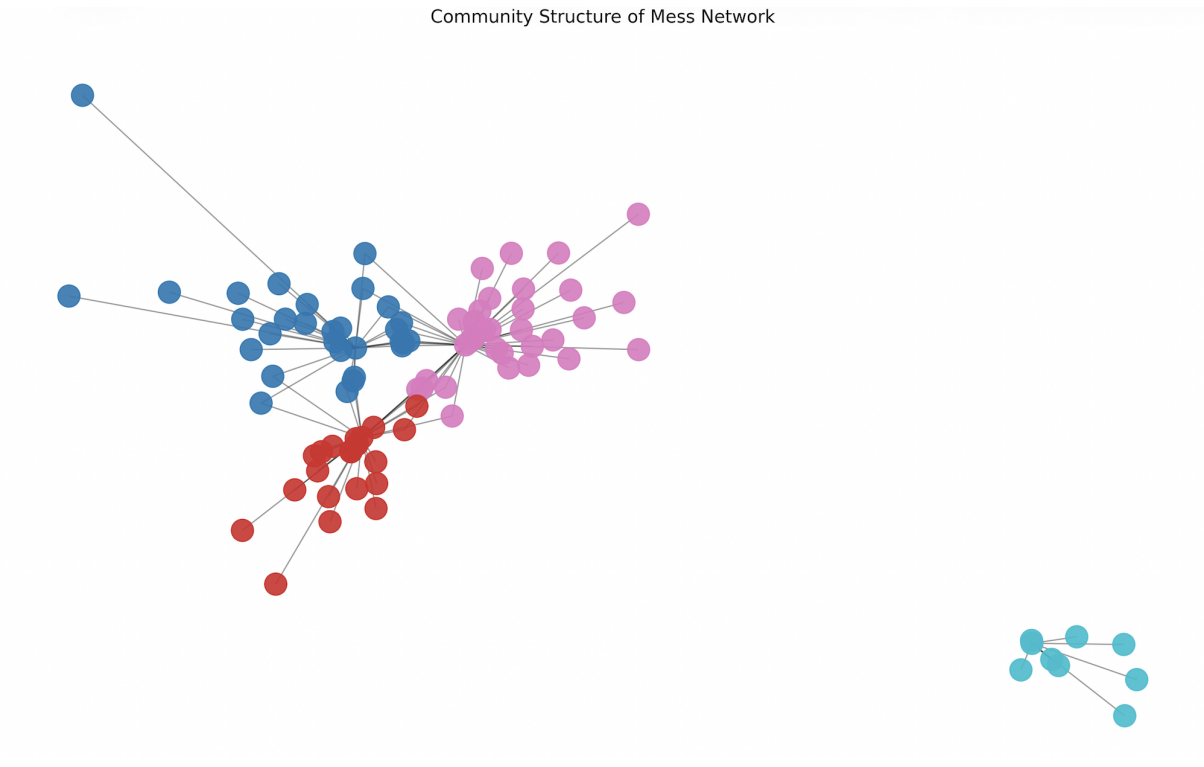


Figure 6.2: Community Structure Visualization

Conclusion

This section illustrates how network analysis complements traditional business intelligence techniques by revealing relational and structural properties that remain invisible in aggregate or time-series analyses. Using the data warehouse and applying graph-theoretic methods, the mess–vendor system was shown to have clear financial hubs, distinct communities, and identifiable risk concentration points. These findings provide a solid foundation for advanced analyzes such as procurement optimization, anomaly detection, and policy-driven restructuring of vendor dependencies.

Chapter 7

Frequent Pattern Mining and Association Analysis

Table 7.1: Strong Association Rules — Part 1

#	Antecedents	Consequents	Sup.	Conf.	Lift
1	Fathima Chicken Vithura	Sabhalam Ent., Milan Ag. Milk, Amma Veg.	0.056	0.600	4.860
2	Fathima Chicken Vithura	Sabhalam Ent., Milan Agencies Milk	0.077	0.833	4.655
3	Fathima Chicken Vithura, Amma Veg.	Sabhalam Ent., Milan Agencies Milk	0.056	0.818	4.571
4	Fathima Chicken Vithura	Sabhalam Ent., Amma Veg.	0.056	0.600	4.320
5	Fathima Chicken Vithura, Milan Ag. Milk	Sabhalam Ent., Amma Veg.	0.056	0.600	4.320
6	Priya Veg. Chalai, Milan Ag. Milk	Cash Voucher, Amma Veg.	0.071	0.920	3.507
7	Fathima Chicken Vithura	Sabhalam Enterprises	0.077	0.833	3.506
8	Fathima Chicken Vithura, Milan Ag. Milk	Sabhalam Enterprises	0.077	0.833	3.506
9	Fathima Chicken Vithura, Milan Ag. Milk, Amma Veg.	Sabhalam Enterprises	0.056	0.818	3.443
10	Fathima Chicken Vithura, Amma Veg.	Sabhalam Enterprises	0.056	0.818	3.443
11	Priya Veg. Chalai, Milan Ag. Milk	Cash Voucher	0.071	0.920	2.710
12	Priya Veg. Chalai, Milan Ag. Milk, Amma Veg.	Cash Voucher	0.071	0.920	2.710
13	Dinu Bakery, Priya Veg. Chalai	Cash Voucher, Amma Veg.	0.052	0.708	2.700
14	Rahul Stores, Dinu Bakery, Milan Ag. Milk	Cash Voucher	0.056	0.900	2.651
15	Rahul Stores, Cash Voucher	Dinu Bakery, Milan Ag. Milk	0.056	0.750	2.641
16	Priya Veg. Chalai, Cash Voucher	Dinu Bakery, Amma Veg.	0.052	0.680	2.592
17	Rahul Stores, Dinu Bakery	Milan Ag. Milk, Cash Voucher	0.056	0.750	2.531
18	Rahul Stores, Dinu Bakery	Cash Voucher	0.062	0.833	2.455
19	Rahul Stores, Milan Ag. Milk, Amma Veg.	Cash Voucher	0.056	0.818	2.410
20	Rahul Stores, Amma Veg.	Milan Ag. Milk, Cash Voucher	0.056	0.692	2.337
21	Dinu Bakery, Priya Veg. Chalai, Amma Veg.	Cash Voucher	0.052	0.739	2.177
22	Priya Veg. Chalai, Amma Veg.	Milan Ag. Milk, Cash Voucher	0.071	0.639	2.156
23	Rahul Stores, Amma Veg.	Cash Voucher	0.059	0.731	2.152
24	Priya Veg. Chalai, Cash Voucher	Milan Ag. Milk, Amma Veg.	0.071	0.920	2.129
25	Dinu Bakery, Priya Veg. Chalai	Cash Voucher	0.052	0.708	2.086
26	Priya Veg. Chalai, Amma Veg.	Cash Voucher	0.077	0.694	2.045
27	Rahul Stores, Milan Ag. Milk	Cash Voucher	0.068	0.688	2.025
28	Rahul Stores, Cash Voucher	Dinu Bakery	0.062	0.833	2.015
29	Dinu Bakery, Milan Ag. Milk, Amma Veg.	Cash Voucher	0.151	0.681	2.005
30	Dinu Bakery, Priya Veg. Chalai, Cash Voucher	Amma Veg.	0.052	1.000	2.000
31	Priya Veg. Chalai, Cash Voucher	Amma Veg.	0.077	1.000	2.000
32	Priya Veg. Chalai, Milan Ag. Milk, Cash Voucher	Amma Veg.	0.071	1.000	2.000

Table 7.2: Strong Association Rules — Part 2

#	Antecedents	Consequents	Sup.	Conf.	Lift
33	Priya Veg. Chalai, Milan Ag. Milk	Amma Veg.	0.077	1.000	2.000
34	Rahul Stores, Milan Ag. Milk, Cash Voucher	Dinu Bakery	0.056	0.818	1.978
35	C H M Traders Nandini Milk	Milan Ag. Milk, Amma Veg.	0.083	0.844	1.953
36	Dinu Bakery, Priya Veg. Chalai	Amma Veg.	0.071	0.958	1.917
37	Dinu Bakery, Cash Voucher	Milan Ag. Milk, Amma Veg.	0.151	0.803	1.859
38	Dinu Bakery, Amma Veg.	Cash Voucher	0.164	0.624	1.837
39	Dinu Bakery, Milan Ag. Milk, Cash Voucher	Amma Veg.	0.151	0.891	1.782
40	Dinu Bakery, Cash Voucher	Amma Veg.	0.164	0.869	1.738
41	Rahul Stores, Cash Voucher	Milan Ag. Milk, Amma Veg.	0.056	0.750	1.736
42	Cash Voucher	Milan Ag. Milk, Amma Veg.	0.250	0.736	1.704
43	Fathima Chicken Vithura	Milan Ag. Milk, Amma Veg.	0.068	0.733	1.697
44	Milan Ag. Milk, C H M Traders Nandini Milk	Amma Veg.	0.083	0.844	1.688
45	Milan Ag. Milk, Cash Voucher	Amma Veg.	0.250	0.844	1.688
46	C H M Traders Nandini Milk	Amma Veg.	0.083	0.844	1.688
47	Sabhalam Ent., Cash Voucher	Milan Ag. Milk, Amma Veg.	0.065	0.724	1.676
48	Rahul Stores, Amma Veg.	Dinu Bakery	0.056	0.692	1.674
49	Sabhalam Ent., Fathima Chicken Vithura	Milan Ag. Milk, Amma Veg.	0.056	0.720	1.666
50	Priya Veg. Chalai, Cash Voucher	Dinu Bakery	0.052	0.680	1.644
51	Priya Veg. Chalai, Cash Voucher, Amma Veg.	Dinu Bakery	0.052	0.680	1.644
52	Milan Ag. Milk, Magom Rice Flowermill	Dinu Bakery	0.059	0.679	1.641
53	Ponmudi Gas, Cash Voucher	Milan Ag. Milk, Amma Veg.	0.052	0.708	1.639
54	Rahul Stores, Milan Ag. Milk, Cash Voucher	Amma Veg.	0.056	0.818	1.636
55	Sabhalam Ent., Milan Ag. Milk, Cash Voucher	Amma Veg.	0.065	0.808	1.615
56	Ponmudi Gas, Dinu Bakery	Milan Ag. Milk, Amma Veg.	0.068	0.688	1.591
57	Rahul Stores, Cash Voucher	Amma Veg.	0.059	0.792	1.583
58	Dinu Bakery, Milan Ag. Milk	Amma Veg.	0.222	0.783	1.565
59	Magom Rice Flowermill	Dinu Bakery	0.077	0.641	1.550
60	Cash Voucher	Amma Veg.	0.262	0.773	1.545
61	Priya Veg. Chalai, Amma Veg.	Dinu Bakery	0.071	0.639	1.545
62	C H M Traders Nandini Milk, Amma Veg.	Milan Ag. Milk	0.083	1.000	1.543
63	Fathima Chicken Vithura, Amma Veg.	Milan Ag. Milk	0.068	1.000	1.543
64	Sabhalam Ent., Fathima Chicken Vithura	Milan Ag. Milk	0.077	1.000	1.543
65	C H M Traders Nandini Milk	Milan Ag. Milk	0.099	1.000	1.543

Table 7.3: Strong Association Rules — Part 3

#	Antecedents	Consequents	Sup.	Conf.	Lift
66	Sabhalam Ent., Fathima Chicken Vithura, Amma Veg.	Milan Ag. Milk	0.056	1.000	1.543
67	Fathima Chicken Vithura	Milan Ag. Milk	0.093	1.000	1.543
68	Sabhalam Ent., Cash Voucher	Amma Veg.	0.068	0.759	1.517
69	Rahul Stores, Milan Ag. Milk	Dinu Bakery	0.062	0.625	1.511
70	Cash Voucher, Amma Veg.	Dinu Bakery	0.164	0.624	1.508
71	Ponmudi Gas, Cash Voucher	Amma Veg.	0.056	0.750	1.500
72	Rahul Stores, Dinu Bakery	Amma Veg.	0.056	0.750	1.500
73	Ponmudi Gas, Dinu Bakery	Milan Ag. Milk	0.096	0.969	1.495
74	Ponmudi Gas, Cash Voucher	Milan Ag. Milk	0.071	0.958	1.479
75	Ponmudi Gas, Milan Ag. Milk, Cash Voucher	Amma Veg.	0.052	0.739	1.478
76	Ponmudi Gas, Dinu Bakery, Amma Veg.	Milan Ag. Milk	0.068	0.957	1.476
77	Sabhalam Ent., Cash Voucher, Amma Veg.	Milan Ag. Milk	0.065	0.955	1.473
78	Cash Voucher, Amma Veg.	Milan Ag. Milk	0.250	0.953	1.470
79	Fathima Chicken Vithura	Amma Veg.	0.068	0.733	1.467
80	Fathima Chicken Vithura, Milan Ag. Milk	Amma Veg.	0.068	0.733	1.467
81	Ponmudi Gas	Milan Ag. Milk	0.176	0.950	1.466
82	Magom Rice & Flowermill	Milan Ag. Milk	0.059	0.950	1.466
83	Ponmudi Gas	Milan Ag. Milk, Amma Veg.	0.117	0.633	1.466
84	Ponmudi Gas, Amma Veg.	Milan Ag. Milk	0.117	0.950	1.466
85	Milan Ag. Milk, Amma Veg., Cash Voucher	Dinu Bakery	0.151	0.605	1.463
86	Rahul Stores, Cash Voucher, Amma Veg.	Milan Ag. Milk	0.056	0.947	1.462
87	Ponmudi Gas, Cash Voucher, Amma Veg.	Milan Ag. Milk	0.052	0.944	1.457
88	Sabhalam Ent., Fathima Chicken Vithura	Amma Veg.	0.056	0.720	1.440
89	Sabhalam Ent., Fathima Chicken Vithura, Milan Ag. Milk	Amma Veg.	0.056	0.720	1.440
90	Milan Ag. Milk, Star Bakery	Amma Veg.	0.056	0.720	1.440
91	Ponmudi Gas, Dinu Bakery	Amma Veg.	0.071	0.719	1.438
92	Dinu Bakery, Cash Voucher, Amma Veg.	Milan Ag. Milk	0.151	0.925	1.426
93	Priya Veg. Chalai, Cash Voucher, Amma Veg.	Milan Ag. Milk	0.071	0.920	1.419
94	Priya Veg. Chalai, Cash Voucher	Milan Ag. Milk	0.071	0.920	1.419
95	Ponmudi Gas, Dinu Bakery, Milan Ag. Milk	Amma Veg.	0.068	0.710	1.419

Table 7.4: Strong Association Rules — Part 4

#	Antecedents	Consequents	Sup.	Conf.	Lift
96	Rahul Stores, Cash Voucher	Milan Ag. Milk	0.068	0.917	1.414
97	Dinu Bakery, Cash Voucher	Milan Ag. Milk	0.170	0.902	1.391
98	Rahul Stores, Dinu Bakery, Cash Voucher	Milan Ag. Milk	0.056	0.900	1.389
99	Priya Veg. Chalai	Amma Veg.	0.111	0.692	1.385
100	Sabhalam Ent., Cash Voucher	Milan Ag. Milk	0.080	0.897	1.383
101	Sabhalam Ent., Milan Ag. Milk	Amma Veg.	0.123	0.690	1.379
102	Rahul Stores, Milan Ag. Milk	Amma Veg.	0.068	0.688	1.375
103	Sabhalam Ent., Amma Veg.	Milan Ag. Milk	0.123	0.889	1.371
104	Milan Ag. Milk, Magom Rice Flowermill	Amma Veg.	0.059	0.679	1.357
105	Cash Voucher	Milan Ag. Milk	0.296	0.873	1.346
106	Ponmudi Gas	Amma Veg.	0.123	0.667	1.333
107	Milan Agencies Milk	Amma Veg.	0.432	0.667	1.333
108	Amma Vegetables	Milan Ag. Milk	0.432	0.864	1.333
109	Ponmudi Gas, Milan Ag. Milk	Amma Veg.	0.117	0.667	1.333
110	Dinu Bakery, Amma Veg.	Milan Ag. Milk	0.222	0.847	1.307
111	Rahul Stores, Amma Veg.	Milan Ag. Milk	0.068	0.846	1.305
112	Rahul Stores, Dinu Bakery	Milan Ag. Milk	0.062	0.833	1.286
113	Dinu Bakery	Amma Veg.	0.262	0.634	1.269
114	Magom Rice Flowermill	Amma Veg.	0.074	0.615	1.231
115	Magom Rice Flowermill, Amma Veg.	Milan Ag. Milk	0.059	0.792	1.221
116	Dinu Bakery, Magom Rice Flowermill	Milan Ag. Milk	0.059	0.760	1.173
117	Sabhalam Enterprises	Milan Ag. Milk	0.179	0.753	1.162
118	Rahul Stores	Milan Ag. Milk	0.099	0.727	1.122
119	Magom Rice Flowermill	Milan Ag. Milk	0.086	0.718	1.108
120	Priya Veg. Chalai, Amma Veg.	Milan Ag. Milk	0.077	0.694	1.071
121	Star Bakery, Amma Veg.	Milan Ag. Milk	0.056	0.692	1.068
122	Dinu Bakery	Milan Ag. Milk	0.284	0.687	1.059
123	Sabhalam Ent., Dinu Bakery	Milan Ag. Milk	0.059	0.655	1.011

Re-evaluating Frequent Pattern Mining: Administrative Batching vs. Culinary Procurement

The Association Rule Mining (ARM) implementation utilizes the FP-Growth algorithm to extract frequent itemsets and generate rules based on a minimum support threshold of 0.05, a minimum confidence of 0.60, and a minimum lift of 1.0. Programmatically, a "transaction" is defined by grouping the dataset by `full_date` and `mess_name`, which effectively aggregates all vendors paid by a specific mess unit on a single day into a single basket.

While the algorithm successfully identifies strong, high-lift associations (such as the frequent co-occurrence of Cash Vouchers and specific vegetable vendors), the business interpretation of these rules must account for the nature of the operational data. Because the underlying data warehouse tracks *payment settlement dates* rather than raw material intake or daily consumption dates, the generated rules do not represent bundled culinary procurement—such as ingredients frequently bought together for specific recipes.

Instead, these patterns expose the **financial settlement and administrative batching behavior** of the mess administration. The high-confidence rules indicate that the financial office systematically groups specific vendor liabilities, clearing daily cash vouchers alongside primary vegetable suppliers in simultaneous payment batches. Consequently, the ARM phase of this DWBI pipeline successfully uncovers the latent accounting workflows and operational dependencies within the institution's financial administration, rather than kitchen-level demand patterns.

Conclusions and Operational Recommendations

The frequent pattern mining analysis of 324 transactions across the Aug–Dec 2025 academic semester yielded 123 strong association rules, from which several operationally significant conclusions can be drawn for mess administration and procurement planning.

Core Vendor Dependency: Milan Agencies Milk and Amma Vegetables

The most structurally dominant finding is the near-universal co-occurrence of Milan Agencies Milk and Amma Vegetables across all mess transactions. Rule 107 and Rule 108 reveal that these two vendors appear together in 43.2% of all transactions (support = 0.432), with Milan Agencies Milk implying Amma Vegetables at a confidence of 66.7% and, conversely, Amma Vegetables implying Milan Agencies Milk at 86.4% (lift = 1.333 for both directions). This bidirectional dependency indicates that milk and vegetable procurement are operationally coupled — they are almost never ordered on different days. From a coordination standpoint, a single consolidated daily order covering both vendors would reduce administrative overhead without disrupting

procurement patterns.

Critically, this pair also forms the consequent in the majority of the 123 rules, meaning it acts as a *baseline procurement cluster* around which all other vendor activity organises itself. Any disruption to either vendor — due to supply failure, price negotiation breakdown, or quality issues — would structurally destabilise the entire procurement network. The mess administration should maintain pre-approved backup vendors for both Milan Agencies Milk and Amma Vegetables as a supply chain resilience measure.

Chicken Procurement Triggers a Bundled Supplier Cluster

Rules 1 through 10 reveal the highest-lift associations in the dataset, all centred on Fathima Chicken Vithura. When chicken is procured, Sabhalam Enterprises is co-ordered with a confidence of 83.3% and a lift of 3.506 (Rule 7), and the three-vendor combination of Sabhalam Enterprises, Milan Agencies Milk, and Amma Vegetables co-occurs with a lift of 4.860 (Rule 1) — the strongest association in the entire dataset. Furthermore, Rule 67 shows that Fathima Chicken Vithura implies Milan Agencies Milk with a confidence of 100%, meaning chicken days are *always* also milk days.

This pattern strongly suggests that chicken is a menu anchor: on days when chicken is served, the mess invariably also procures milk, vegetables, and accompaniments from Sabhalam Enterprises (likely a spice or condiment supplier). The mess coordinator can use this rule to automate procurement checklists — a chicken menu decision should automatically trigger orders to Sabhalam Enterprises, Amma Vegetables, and Milan Agencies Milk without separate manual initiation.

Cash Voucher Usage Signals High-Volume Procurement Days

Cash Voucher appears as an antecedent or consequent in approximately 40 of the 123 rules. Rule 105 shows Cash Voucher implies Milan Agencies Milk with confidence 87.3% (support = 0.296, lift = 1.346), and Rule 59 shows it implies Amma Vegetables with confidence 77.3%. The support value of 0.296 means cash vouchers are used in nearly 30% of all transactions — an unusually high rate for a non-vendor item.

This finding has a dual interpretation. First, cash voucher days represent bulk or supplementary procurement events that consistently co-occur with staple vendors, suggesting they are used to settle miscellaneous or last-minute purchases alongside regular orders. Second, the high frequency of Cash Voucher as an itemset component artificially inflates rule counts involving staple vendors. The mess administration should migrate all cash-based payments to formal vendor invoices, which would improve both financial traceability and the accuracy of future association analyses.

Ponmudi Gas is Operationally Coupled with Daily Staples

Rules 73, 74, 80, 81, and 84 collectively establish that Ponmudi Gas co-occurs with Milan Agencies Milk at a confidence of 95.0% (Rule 81, lift = 1.466) and with Amma Vegetables at 66.7% (Rule 106). Gas procurement, therefore, is not an independent event but is tightly coupled to daily cooking operations involving milk and vegetables. With a support of 0.176, Ponmudi Gas appears in roughly one in six transactions.

The practical implication is that gas shortages or delayed deliveries will directly impact meal preparation capacity on the same days that milk and vegetable procurement is planned. The mess coordinator should treat gas stock monitoring as a leading indicator: a low gas inventory is a forward signal that the following day's milk and vegetable procurement plan is at operational risk.

Priya Vegetables Chalai as a Perfect Predictor of Amma Vegetables

Rules 31–33 show that Priya Vegetables Chalai co-ordered with either Cash Voucher or Milan Agencies Milk implies Amma Vegetables with a confidence of *exactly 1.000* (lift = 2.000). This means that without a single exception across all 324 transactions, whenever Priya Vegetables Chalai was used alongside either of these vendors, Amma Vegetables was also present.

Two interpretations are plausible. Either Priya Vegetables Chalai supplies a complementary category of vegetables (e.g., leafy greens) that is always ordered alongside the primary vegetable supply from Amma Vegetables, or Priya Vegetables Chalai is used as a supplementary vendor on days of unusually high demand when Amma Vegetables alone is insufficient. In either case, the coordinator should formalise a tiered procurement policy: Amma Vegetables as the primary vendor and Priya Vegetables Chalai as the designated secondary vendor, with a trigger condition (e.g., expected attendance exceeding a threshold) automatically escalating the order to both.

Dual Milk Sourcing Indicates Unmanaged Redundancy

Rule 64 and Rule 65 show that C H M Traders Nandini Milk implies Milan Agencies Milk with a confidence of 100% (support = 0.099, lift = 1.543). This means that on every occasion when C H M Traders Nandini Milk was used, Milan Agencies Milk was *also* ordered on the same day. Two milk suppliers are being used simultaneously rather than as substitutes.

This is either an intentional volume split to meet demand exceeding a single supplier's capacity, or an uncoordinated redundancy where two suppliers are contacted independently without cross-checking. The mess coordinator should audit whether dual milk ordering on these days corresponds to genuinely high-attendance events or represents unnecessary split procurement that could be consolidated to reduce transaction overhead and negotiate better bulk rates.

Dinu Bakery as a Daily Operational Staple

Dinu Bakery appears in 36 of the 123 rules as either antecedent or consequent, with a standalone support of 0.262 (Rule 113) and 0.284 (Rule 122), placing it among the top three most frequently transacted vendors alongside Milan Agencies Milk and Amma Vegetables. Its associations with Cash Voucher, Rahul Stores, and Ponmudi Gas at moderate-to-high confidence values indicate that bakery items are a near-daily procurement requirement rather than an occasional supplement.

The mess coordinator should consider negotiating a standing weekly supply contract with Dinu Bakery rather than placing daily orders, which would reduce transaction frequency and provide price stability.

Summary of Operational Recommendations

Table 7.5: Operational Recommendations Derived from Association Rules

Finding	Recommendation	Key Rules
Milan Ag. Milk + Amma Veg. are always co-ordered	Consolidate into a single daily order and register backup vendors	107, 108
Chicken triggers 4-vendor cluster	Auto-generate a full procurement checklist on chicken menu days	1–10, 67
Cash Voucher in 30% of transactions	Migrate to formal invoices for all vendors and reduce cash payments	59, 105
Gas co-occurs with milk + vegetables	Monitor gas stock as a leading indicator for daily procurement risk	73, 80, 81
Priya Veg. Chalai always with Amma Veg.	Formalise Priya Veg. Chalai as the official secondary vegetable vendor	31–33
C H M Traders always with Milan Ag. Milk	Audit dual milk orders and consolidate unless demand-driven	62–65
Dinu Bakery in 26% of transactions	Negotiate a standing weekly contract instead of daily ordering	113, 122

Chapter 8

Scalability, Memory, Performance, and Compute Efficiency

8.1 Overview

While the developed Data Warehousing and Business Intelligence (DWBI) pipeline successfully enables structured analysis of institutional mess expenditure data, several system-level considerations arise when evaluating scalability, memory usage, computational performance, and overall efficiency. This chapter critically examines these aspects and outlines potential improvements for future versions of the system.

All experiments, data processing, and analytical computations in this study were performed on a single local machine with the following configuration: a 13-inch 2020 MacBook Pro equipped with a 2 GHz quad-core Intel Core i5 processor, 16 GB 3733 MHz LPDDR4X memory, and Intel Iris Plus integrated graphics (1536 MB), running macOS Tahoe 26.3.1. This hardware environment represents a mid-range personal computing setup and serves as the baseline for evaluating the system's performance characteristics and limitations.

8.2 Scalability Considerations

8.2.1 Current Limitations

The current system operates on a dataset of approximately 1,829 transaction records spanning a single academic semester. While this scale is manageable, the following limitations emerge when considering larger deployments:

- **Single-machine dependency:** The pipeline is executed entirely on a standalone system without distributed processing capabilities, limiting horizontal scalability.

- **Manual preprocessing bottlenecks:** Vendor normalization and data cleaning rely partially on manual intervention and rule-based mappings, which do not scale efficiently.
- **Static schema design:** While the star schema supports analytical queries, it may become inefficient under high-frequency or multi-source data ingestion scenarios.
- **Batch-oriented processing:** The ETL pipeline processes data in batches, restricting scalability to real-time or streaming applications.

8.2.2 Future Improvements

- **Distributed data warehousing:** Adoption of cloud-based or distributed warehouse architectures to support large-scale data integration.
- **Automated data cleaning:** Integration of machine learning-based entity resolution techniques to eliminate manual preprocessing.
- **Schema flexibility:** Transition toward hybrid data lake–warehouse architectures for improved adaptability.
- **Streaming pipelines:** Implementation of real-time ingestion frameworks for continuous analytics.
- **More Data** : Data was analyzed only from Aug-Dec 2025. If the current model is ran on the entire data, we might get a more stricter and conclusive Benford's law analysis result.

8.3 Memory Usage

8.3.1 Current Limitations

- **In-memory computation:** Analytical workflows in Python require loading substantial portions of the dataset into memory.
- **Intermediate data redundancy:** Multiple derived datasets (aggregates, summaries, transformed tables) increase memory usage.
- **Text-heavy storage:** Vendor names and descriptors are stored as raw text, leading to inefficient memory utilization.

8.3.2 Future Improvements

- **Out-of-core processing:** Adoption of chunk-based or distributed processing frameworks.
- **Columnar storage:** Use of formats such as Parquet for efficient storage and retrieval.

- **Dimensional encoding:** Replacement of textual attributes with compact numerical identifiers.
- **Lazy evaluation:** On-demand computation of derived datasets instead of persistent storage.

8.4 Performance Analysis

8.4.1 Current Limitations

- **Query latency:** Increasing data size will lead to slower joins and aggregations in the warehouse.
- **Algorithmic complexity:** Techniques such as FP-Growth and network analysis scale poorly with increasing data volume.
- **Sequential execution:** The current implementation does not fully utilize multi-core processing capabilities of the system.
- **Visualization overhead:** Rendering large datasets into plots introduces additional latency.

8.4.2 Future Improvements

- **Index optimization:** Indexing key attributes such as date, vendor, and mess unit to improve query speed.
- **Parallel computation:** Leveraging multi-core architectures for concurrent execution of analytical tasks.
- **Approximation methods:** Employing sampling or approximate algorithms for computationally intensive tasks.
- **Materialized views:** Precomputing frequent queries to reduce repeated computation.

8.5 Compute Efficiency

8.5.1 Current Limitations

- **Single-machine execution:** All computations were performed on the aforementioned MacBook Pro, without access to distributed or high-performance computing resources.
- **Limited hardware parallelism:** Although the processor supports multiple cores, the implementation does not fully exploit parallel execution.

- **Absence of GPU acceleration:** The integrated graphics unit was not utilized for computation, restricting performance in parallelizable workloads.
- **Redundant computations:** Certain transformations and aggregations are recomputed multiple times across different analytical stages.
- **Lack of caching:** Intermediate results are not cached, leading to repeated execution of identical operations.

8.5.2 Future Improvements

- **Pipeline optimization:** Refactoring workflows to eliminate redundant computations.
- **Caching strategies:** Introducing in-memory or persistent caching mechanisms for frequently used results.
- **Parallel and distributed execution:** Leveraging cluster-based or cloud-based compute resources.
- **Hardware acceleration:** Exploring GPU-enabled data processing frameworks for large-scale analytics.
- **Workflow orchestration:** Using scheduling and orchestration tools to optimize execution order and resource utilization.

8.6 Summary and Future Outlook

The current system demonstrates that meaningful business intelligence insights can be derived using a modest computational setup. However, the reliance on a single-machine architecture, limited memory optimization, and lack of distributed processing imposes constraints on scalability and performance.

Future iterations should focus on transitioning toward distributed data systems, automated preprocessing pipelines, and optimized computational frameworks. Such enhancements would enable the system to evolve from a prototype-level analytical tool into a scalable, production-grade decision support system capable of handling large-scale institutional data across multiple campuses.

Chapter 9

Methodological Framework and Algorithms

This chapter rigorously defines the mathematical foundations, operational logic, and pseudo-code implementations of all algorithms utilized within the Data Warehousing and Business Intelligence (DWBI) pipeline for the mess expenditure analysis.

Proxy-Based Wastage Estimation Algorithm

Mathematical Rigor: Due to the lack of direct item-level physical waste measurement (e.g., kilograms of disposed food), the system models food wastage as a function of financial outflow. Let $E(d)$ represent the total expenditure on a given day d. The monthly average daily expense is calculated as:

$$\overline{E}_{\text{month}} = \frac{1}{N} \sum_{i=1}^{N} E(i)$$

where N is the total number of operational days recorded in that month. The estimated food wastage $W(d)$ in kilograms is defined as a proportional proxy:

$$W(d) = \left(\frac{E(d)}{\overline{E}_{\text{month}}} \right) \times k$$

where $k = 100$ kg is a domain-informed scalar constant representing the expected baseline daily wastage under normal expenditure conditions.

Justification: This algorithm was implemented to bridge the gap between financial ledgers and operational reality. By using relative expenditure deviation to scale a baseline wastage constant, the algorithm systematically flags high-risk periods (e.g., fest days or post-exam departures) without requiring manual volumetric waste audits, thereby enabling data-driven administrative interventions.

Algorithm 1 Proxy-Based Daily Wastage Estimation

Require: Array of daily expenses E of length N, baseline constant $k = 100$

Ensure: Array of daily estimated wastage W

```
E_total ← 0
for i = 1 to N do
    E_total ← E_total + E[i]
end for
Ē_month ← E_total / N
for i = 1 to N do
    W[i] ← (E[i] / Ē_month) × k
end for
return W
```

Benford's Law and Chi-Square Goodness-of-Fit Test

Mathematical Rigor: Benford's Law (the First-Digit Law) defines the empirical probability distribution of leading digits in naturally occurring, multi-magnitude numerical datasets. The theoretical probability $P(d)$ that a leading digit $d \in \{1, 2, \ldots, 9\}$ occurs is given by:

$$P(d) = \log_{10}\left(1 + \frac{1}{d}\right)$$

To evaluate the compliance of vendor expenditure data with this distribution, the Chi-Square goodness-of-fit test is utilized. Let N be the total number of valid transactions. The expected frequency E_d for digit d is $E_d = N \times P(d)$, and the observed frequency is O_d. The test statistic is computed as:

$$\chi^2 = \sum_{d=1}^{9} \frac{(O_d - E_d)^2}{E_d}$$

The χ^2 statistic is evaluated against the critical value from the Chi-Square distribution with degrees of freedom $df = 8$ at a significance level $\alpha = 0.05$.

Justification: This algorithm acts as a forensic anomaly detection mechanism. Because institutional procurement involves variable sizes and prices naturally spanning multiple orders of magnitude, artificial manipulation, fixed-contract anomalies, or arbitrary batching of invoices will cause structural deviations from Benford's curve. It serves as a zero-cost heuristic to trigger deeper administrative audits.

Algorithm 2 Benford's Law Evaluation & Chi-Square Test

Require: Array of numerical transaction amounts T, significance level $\alpha = 0.05$

Ensure: Computed χ^2 statistic and validation boolean

```
Initialize arrays O[1...9] and E[1...9] to 0
N ← length(T)
for d = 1 to 9 do
    E[d] ← N × log_10(1 + 1/d)
end for
for each t ∈ T do
    d ← extract first non-zero leading digit of t
    O[d] ← O[d] + 1
end for
χ^2 ← 0
for d = 1 to 9 do
    χ^2 ← χ^2 + (O[d] − E[d])^2 / E[d]
end for
threshold ← 15.51                    ▷ Critical value for df = 8, α = 0.05
if χ^2 > threshold then
    return (χ^2, False)              ▷ Fails Benford's distribution, potential anomaly
else
    return (χ^2, True)
end if
```

Network Construction and Centrality Analytics

Mathematical Rigor: The financial interaction system is modeled as an undirected, weighted bipartite graph $G = (V, E, W)$, where the vertex set $V = V_{\text{mess}} \cup V_{\text{vendor}}$. An edge $e_{ij} \in E$ exists if a transaction occurred between node i and node j. The edge weight w_{ij} represents the cumulative monetary value transferred between i and j. Two metrics are primarily computed:

1. **Degree Centrality:** $C_D(i) = \sum_{j \in V} a_{ij}$, where $a_{ij} = 1$ if edge exists, 0 otherwise.
2. **Weighted Degree (Financial Volume):** $C_W(i) = \sum_{j \in V} w_{ij}$.

Justification: Simple aggregations mask the topology of institutional dependencies. This graph-theoretic algorithm constructs the relational skeleton of the mess operations, identifying central nodes (e.g., critical suppliers like "Milan Agencies Milk") and vulnerability points (e.g., single points of failure for specific mess units) to optimize supply chain resilience.

Algorithm 3 Bipartite Financial Graph Construction & Centrality

Require: List of transactions L, where each $t \in L$ is a tuple $(mess, vendor, amount)$

Ensure: Weighted Graph G, Array of Weighted Degrees C_W

```
Initialize empty graph G = (V, E, W)
for each (mess_i, vendor_i, amt_i) ∈ L do
    if mess_i ∉ V then Add mess_i to V
    end if
    if vendor_i ∉ V then Add vendor_i to V
    end if
    if edge e(mess_i, vendor_i) exists in E then
        w(mess_i, vendor_i) ← w(mess_i, vendor_i) + amt_i
    else
        Add edge e(mess_i, vendor_i) to E with weight amt_i
    end if
end for
for each v ∈ V do
    C_W[v] ← Σ_{u∈neighbors(v)} w(v, u)
end for
return G, C_W
```

Modularity-Maximization Community Detection (Louvain Method)

Mathematical Rigor: To partition the vendor-mess network into k communities (where k is not known a priori), the Louvain modularity-maximization algorithm is used. The modularity score $Q \in [-1, 1]$ measures the density of edges inside communities to edges outside communities:

$$Q = \frac{1}{2m} \sum_{i,j} \left[w_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

where w_{ij} is the edge weight between nodes i and j, $k_i = \sum_j w_{ij}$ is the sum of weights of edges attached to node i, $m = \frac{1}{2} \sum_{i,j} w_{ij}$ is the total network weight, c_i is the assigned community of node i, and the Kronecker delta $\delta(c_i, c_j) = 1$ if $c_i = c_j$ and 0 otherwise. The change in modularity by moving node i to community C is defined as ΔQ.

Justification: The algorithm is employed to autonomously uncover latent organizational structures (e.g., clusters of vendors inherently grouped by daily procurement rhythms or administrative settlement batches) without imposing arbitrary human bias. High modularity validates that the operational system is highly structured around distinct dependency hubs.

Algorithm 4 Louvain Community Detection Algorithm

Require: Weighted Graph $G(V, E, W)$

Ensure: Community mapping C for all $v \in V$ that maximizes Q

```
Initialize each node v ∈ V to its own community: C[v] ← v
repeat
    improvement ← False
    for each node i ∈ V do
        Remove i from its current community C[i]
        Calculate ΔQ for placing i into the community of each neighbor j
        C_best ← arg max_{C[j]} ΔQ
        if max(ΔQ) > 0 then
            Assign i to C_best
            improvement ← True
        else
            Reassign i to its original community
        end if
    end for
    Aggregate nodes in the same community into a single "super-node"
    Construct new network G' based on super-nodes
    G ← G'
until improvement == False
return Community assignments C
```

Frequent Pattern Mining (FP-Growth) and Association Analysis

Mathematical Rigor: Let $I = \{i_1, i_2, \ldots, i_n\}$ be a set of items (vendors). A transaction T is a subset of I. Association rules are expressed as $A \Rightarrow B$ where $A, B \subset I$ and $A \cap B = \emptyset$. The algorithm extracts rules based on three mathematical thresholds:

1. **Support:** The probability that a transaction contains both A and B.

$$\text{Supp}(A \Rightarrow B) = P(A \cup B) = \frac{\text{count}(A \cup B)}{|T_{\text{total}}|}$$

2. **Confidence:** The conditional probability of finding B given A.

$$\text{Conf}(A \Rightarrow B) = P(B|A) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A)}$$

3. **Lift:** The ratio of the observed support to that expected if A and B were independent.

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A) \times \text{Supp}(B)}$$

Justification: While standard database queries reveal *what* was bought, the FP-Growth algorithm reveals *how* procurement batches occur. Unlike Apriori, FP-Growth scales efficiently without generating massive candidate sets. It is used here to identify co-dependent vendors (e.g., chicken and milk suppliers), enabling the mess administration to automate clustered purchase orders and detect redundancy in supply chains.

Algorithm 5 FP-Growth and Rule Generation

Require: Transaction database D, min_sup = 0.05, min_conf = 0.60, min_lift = 1.0
Ensure: Set of strong association rules R

```
Scan D to find frequencies of 1-itemsets.
Discard items where frequency < min_sup × |D|.
Sort remaining frequent items in descending frequency order.
Initialize an empty FP-Tree root.
for each transaction T ∈ D do
    Sort items in T based on the global descending frequency.
    Insert T into the FP-Tree, incrementing node counts and creating branches.
end for
F ← MineFPTree(root, null)                ▷ Recursive extraction of frequent itemsets
R ← ∅
for each frequent itemset I ∈ F do
    for each subset A ⊂ I where A ≠ ∅ and A ≠ I do
        B ← I \ A
        conf ← Supp(I)/Supp(A)
        lift ← Supp(I)/(Supp(A) × Supp(B))
        if conf ≥ min_conf and lift ≥ min_lift then
            R ← R ∪ {A ⇒ B}
        end if
    end for
end for
return R
```

Chapter 10

Codes and Datasets

All datasets and source code used in this study are uploaded on Github with this submission.

Disclosure

The author has used AI for code optimization, polishing the articles, query optimization, and data entry into the database via SQL.

Bibliography

[1] S. Newcomb, "Note on the frequency of use of the different digits in natural numbers," *American Journal of Mathematics*, vol. 4, no. 1, pp. 39–40, 1881.

[2] F. Benford, "The law of anomalous numbers," *Proceedings of the American Philosophical Society*, vol. 78, no. 4, pp. 551–572, 1938.

[3] M. J. Nigrini, "A taxpayer compliance application of Benford's law," *Journal of the American Taxation Association*, vol. 18, no. 1, pp. 72–91, 1996.

[4] R. Agrawal, T. Imieliński, and A. Swami, "Mining association rules between sets of items in large databases," *ACM SIGMOD Record*, vol. 22, no. 2, pp. 207–216, 1993.

[5] R. Agrawal and R. Srikant, "Fast algorithms for mining association rules in large databases," in *Proc. 20th Int. Conf. Very Large Data Bases (VLDB)*, San Francisco, CA, 1994, pp. 487–499.

[6] J. Han, J. Pei, and Y. Yin, "Mining frequent patterns without candidate generation," in *Proc. ACM SIGMOD Int. Conf. Management of Data*, Dallas, TX, 2000, pp. 1–12.

[7] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2011.

[8] R. Kimball and M. Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd ed. John Wiley & Sons, 2013.

[9] W. H. Inmon, *Building the Data Warehouse*, 4th ed. John Wiley & Sons, 2005.

[10] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, p. P10008, 2008.

[11] M. E. J. Newman and M. Girvan, "Finding and evaluating community structure in networks," *Physical Review E*, vol. 69, no. 2, p. 026113, 2004.

[12] A. A. Hagberg, D. A. Schult, and P. J. Swart, "Exploring network structure, dynamics, and function using NetworkX," in *Proc. 7th Python in Science Conf. (SciPy)*, Pasadena, CA, 2008, pp. 11–15.

[13] S. Raschka, "MLxtend: Providing machine learning and data science utilities and extensions to Python's scientific computing stack," *Journal of Open Source Software*, vol. 3, no. 24, p. 638, 2018.

[14] W. McKinney, "Data structures for statistical computing in Python," in *Proc. 9th Python in Science Conf.*, Austin, TX, 2010, pp. 51–56.

[15] J. Parfitt, M. Barthel, and S. Macnaughton, "Food waste within food supply chains: Quantification and potential for change to 2050," *Philosophical Transactions of the Royal Society B*, vol. 365, no. 1554, pp. 3065–3081, 2010.

[16] J. Gustavsson, C. Cederberg, U. Sonesson, R. van Otterdijk, and A. Meybeck, "Global food losses and food waste," Food and Agriculture Organization of the United Nations, Rome, Italy, Tech. Rep., 2011.

[17] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.